



CHAPTER 4

FUNDAMENTALS OF ORGANIZATION STRUCTURE

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Opel España

Opel España is the Spanish subsidiary of the global auto giant General Motors (GM), based near the northern Spanish city of Zaragoza. While US auto manufacturers, including GM, are facing increasingly difficult market conditions in their home base due to competition from Asian car builders, global operations had been performing well, at least until the global downturn that hit manufacturing industry everywhere starting in 2007/2008. One of GM's survival strategies is based on continuing its international growth as it rebalances its activities towards growing markets outside the United States. The Zaragoza plant is GM's second largest in Europe, with over 5400 staff and annual capacity of around 480,000 vehicles, and makes the high-selling Corsa, the Meriva minivan and, since 2014, the Mokka subcompact SUV.

Consumer expectations are becoming increasingly sophisticated and differentiated. In the early days of car manufacture, Henry Ford was famously quoted as saying of his Model T, 'the customer can have any colour he wants so long as it's black'.¹ Nowadays, consumer preferences are key. Car buyers demand fashionable interiors and exteriors, technological innovation, a variety of product options to fit individual needs and high quality at reasonable prices. In addition, competition from other manufacturers has intensified, and customers have a range of high quality, competitively priced alternatives. By 2024, all cars made by Opel will also have an electric version due to the climate requirements of the customers as well as society.

This increasingly demanding business environment is leading car companies to adopt various new production models, including what is termed *agile manufacturing*.² Agile manufacturing departs from the traditional mass production approach, focusing on highly customized manufacturing, producing vehicles as and when customers require them, rapidly changing production to meet new demands, and marketing a wide variety of products and product options to satisfy an increasingly diverse and knowledgeable customer base.

Opel España has spent considerable time and effort tailoring its physical plant, equipment, organizational structure and management systems to meet

these challenges and to allow continued growth and profitability. For the launch of the fourth-generation Opel Corsa D in 2006, managers brought in experts from GM/Opel's International Technical Development Centre in Germany to design more efficient and flexible production. The strategy emphasized the need for integrated improvements. Equipment innovations included expanded use of robots in production and new paint-spraying machinery that produces a higher quality finish and fewer environmental emissions.³ Production processes and procedures were streamlined. Central to the production redesign, however, were improvements in organizational design.

Organizational design is centred around teamwork practices designed to build empowerment, involvement and motivation of employees, from top management to the shop floor. From the beginning, the new Corsa's design was based on the integration of production, engineering and marketing managers in the development team.⁴ This teamwork approach extends into the manufacturing process, where flexible manufacturing systems are based on a modular approach. The modular approach allows the end-customer to select from a wide range of standardized options so that the car that rolls off the production line is the one that the consumer has chosen. Like many other manufacturers, Opel España relies increasingly on the outsourcing of various parts of the vehicle. Again, the company is moving towards modular outsourcing, so that rather than subcontracting numerous small components, suppliers provide whole modules, such as steering assemblies, which are then incorporated into the car at the Zaragoza plant. Major suppliers, too, were involved in the Corsa design process from the earliest stage, ensuring that potential problems were smoothed out before full production began.

GM had originally planned to divest itself of its European plants when the parent company ran into trouble in the mid-2000s, but once the decision was made to keep its European division, it placed an emphasis on quality and efficiency. This became even more important in the wake of the economic crisis, which affected Spain particularly badly. However, the great majority of production at the Zaragoza plant is destined for export, albeit mainly within the Eurozone, which remains in the doldrums a decade after the start of the 2007 global financial crisis. Like many companies, Opel España is looking at China as an export market to compensate for weak domestic markets. Volumes have gradually increased from around 265,000 in 2012 towards the plant's 480,000 capacity, with a significant milestone passed in 2019 when the Zaragoza factory produced its 13 millionth car.⁵

Managers responsible for the design and redesign of organizations have used various structural arrangements with regard to the vertical and horizontal divisions of labour. Nearly every organization undergoes structural reorganization at some point to help meet new challenges or to signal the arrival of a ‘new broom’. Structural changes reflect new strategies or respond to changes which, in Chapter 3, were conceived as ‘contingency factors’, such as environment, technology, size and life cycle, and culture. In general, the effectiveness of a direct, authoritarian management style has been increasingly criticized for its lack of responsiveness and ‘agility’ and there has been increased experimentation with more flexible, teamwork-based approaches in an effort to improve decision-making and put resources in places where they are assessed to produce the greatest value.⁶

Purpose of this Chapter

This chapter introduces basic concepts of organization structure. First we define a structure and provide an overview of structural design. Then, an information-processing perspective explains how information flow may be facilitated by designing vertical and horizontal linkages to achieve information flow. Basic design options are then presented, followed by strategies for grouping organizational activities into simple, functional, divisional, matrix, horizontal, virtual network or hybrid structures. The final section examines how the application of designs depends on the organization’s situation and outlines the symptoms of structural misalignment.

Organization Structure



ONLINE
COUNTERPOINT 4.1

There are three key components in the definition of **organization structure**:

- 1 Organization structure designates formal reporting relationships, including the number of levels in the hierarchy and the span of control of managers and supervisors.
- 2 Organization structure identifies the grouping together of individuals into departments and of departments into the total organization.
- 3 Organization structure includes the design of systems to ensure effective communication, coordination and integration of efforts across departments.⁷

These three elements of structure refer to both vertical and horizontal aspects of organizing. For example, the first two elements are the structural *framework*, which include the vertical hierarchy.⁸ The third element pertains to the pattern of *interactions* among organizational employees.

Organization Structure Represented Visually in Organization Charts



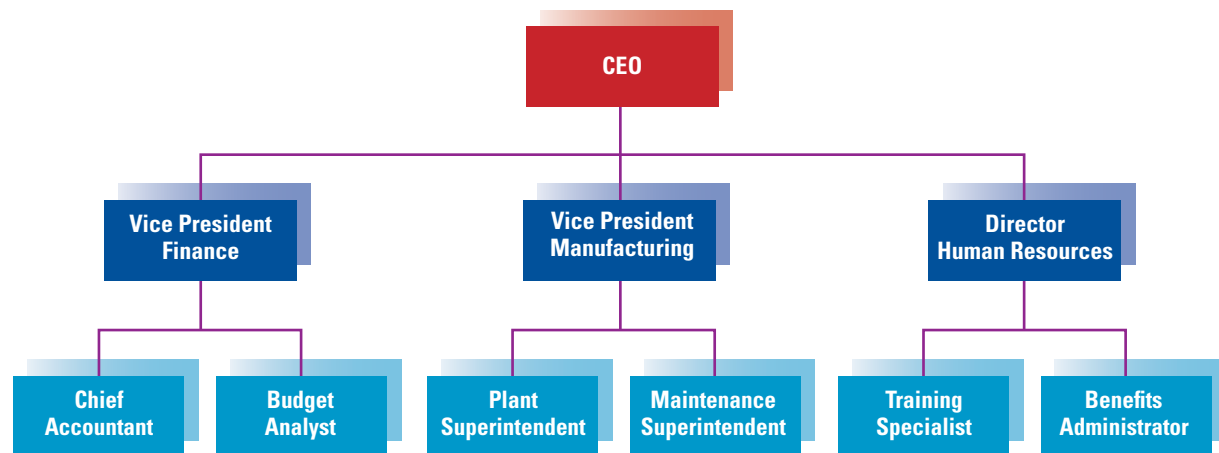
ONLINE BRIEF 4.1

It isn’t possible to see the internal structure of an organization the way we might see its manufacturing tools, offices or products. Although we might see employees going about their duties, performing different tasks and working in different locations, a way to show the structure underlying all this activity is through the organization chart. The organization chart is the (usually top-down) visual representation of a whole set of underlying activities and processes in an organization. Exhibit 4.1 shows a sample organization chart.

The concept of an organization chart, showing what positions exist, how they are grouped and who reports to whom, has been around for centuries.⁹ Diagrams outlining church hierarchy can be found in medieval churches in Spain. However, the contemporary use of the organization chart for business stems from the Industrial Revolution, and especially from the emergence of large-scale steam railways in the mid-nineteenth century. As work grew more complex and was performed by greater and greater numbers of workers, there was a pressing need to develop ways

of managing and controlling organizations (see Counterpoint 4.1). One of the earliest examples of an organizational chart was in US railways, as a response to safety issues. After two passenger trains collided in Massachusetts in 1841, the public demanded better control of the operation. As a result, the Board of Directors of the Western Railroad took steps to outline ‘definite responsibilities for each phase of the company’s business, drawing solid lines of authority and command for the railroad’s administration, maintenance and operation’.¹⁰

EXHIBIT 4.1 A Sample Organization Chart



COUNTERPOINT 4.1

An organization chart is only so many lines and boxes. The organization chart indicates the structure of control, but its enactment and realization depend upon the employees who provide the behaviour. Its purpose is to offer a common frame of reference that, in general and above all, asserts and reinforces a hierarchical relationship of domination between executives at the apex, managers in the middle levels and workers at the base. If all employees can be encouraged to accept this hierarchy as something that is a necessity, rather than politically imposed or expedient, then the goals of the organization determined by executives will more likely be realized through their actions. The chart is not only a technical guideline to encourage people to work together but a powerful political symbol whose acceptance implies deference to its demands.

The type of organization structure that grew out of these efforts to draw lines of command in the late nineteenth and early twentieth centuries was one in which the CEO was placed at the top and everyone else was arranged in layers down below, as illustrated in Exhibit 4.1. It indicates how thinking and decision-making are to be carried out by those at the top, and the physical work is performed by employees who are organized into distinct, functional departments where they are expected to carry out demands. This structure became increasingly entrenched in business, nonprofit and military organizations during the twentieth century. However, such a structure is not always effective, particularly in rapidly changing environments, and a variety of alternative structures have been devised to be more flexible and to increase involvement of all the organization’s employees. These are typically aimed at increasing horizontal coordination and communication and encouraging adaptation to external changes. This chapter’s *Bookmark 4.0* asks whether business is on the verge of a historic transformation, in which traditional forms of organizing are giving way to alternative decentralized, networked structures where more emphasis is placed on horizontal processes.

COUNTERPOINT 4.2

Decentralized structures can improve organizational flexibility and productivity. Employees' experience of decentralized structures, however, may be mixed. As decentralization increases responsibilities of line workers, employees have the chance to develop added competencies. Yet added responsibility and productivity does not necessarily come with improved salaries and job security. Are the benefits of organizational innovation being fairly shared between the company's shareholders and its employees?¹¹ If not, can this continue without stoking up resentment and fuelling resistance in the form of psychological distancing, withdrawal of goodwill, organized opposition, individual sabotage and so on?

BOOKMARK 4.0

Have you read this book?

The Future of Work: How the New Order of Business Will Shape Your Organization, Your Management Style and Your Life

BY THOMAS W. MALONE

Organizations are experiencing tremendous change, and Thomas W. Malone suggests in his book *The Future of Work* that they are on the verge of a fundamental shift that could be 'as important to business as the shift to democracy has been for government'. Rigid, highly centralized vertical hierarchies, he says, will essentially be a thing of the past as organizations move to flexible decentralized forms of organizing based on horizontal work processes. Command-and-control management and top-down decision-making will give way to empowered teams of employees focused on specific workflows and processes, working across organizational boundaries and making their own decisions based on up-to-the-minute information.

The Brave New World of Work

Malone describes several decentralized management structures and provides numerous examples of organizations that are experimenting with new forms of organizing and innovative management techniques. Here are some of Malone's key points about the future of work:

- *Information technology is the key driver of the transformation.* The falling cost of communication

is making the distribution of power away from the corporate suite both inevitable and desirable. Outsourcing information work to India, for example, is possible because digital communication with India is so cheap, as is labour in India. In the same way, accessible information makes it possible for any lower-level employee to plan his or her work more effectively, network and get advice from people anywhere, and make good decisions based on accurate information.

- *Managers will move from command-and-control to coordinate-and-cultivate.* To coordinate is to organize work so that good things happen, whether managers are 'in control' or not. To cultivate means to bring out the best in employees with the right combination of control and freedom. W. L. Gore, the maker of Gore-Tex fabric, lets people decide what they want to do. Leaders emerge based on who has a good idea and can recruit people to work on it. AES, one of the world's largest electric power producers, coordinates and cultivates so well that it lets low-level workers make critical multimillion-dollar decisions about such things as acquiring new subsidiaries.

The Transition Years

Malone, a professor at MIT's Sloan School of Management, acknowledges that hierarchy and centralization continue to provide tremendous advantages for some companies in today's economy. In addition, for most organizations, centralized and decentralized structures and management systems will coexist well into the future. Yet, he is convinced that, ultimately, rigid hierarchical centralized structures will be consigned to the dustbin of history.

The Future of Work: How the New Order of Business Will Shape Your Organization, Your Management Style and Your Life, by Thomas W. Malone, is published by Harvard Business School Press (2004).

Information-Processing Perspective on Structure

Exhibit 4.2 compares organizations designed for control with those designed for learning (see also Chapter 2). An *emphasis on control* is associated with specialized tasks, a hierarchy of authority, rules and regulations, formal reporting systems, few teams or task forces and **centralized** decision-making, which means problems and decisions are funnelled to top levels of the hierarchy for resolution. An *emphasis on learning* is associated with shared tasks, a relaxed hierarchy, few rules, face-to-face communication, many teams and task forces and informal, **decentralized** decision-making. Decentralized decision-making means decision-making authority is pushed down to lower organizational levels (see Counterpoint 4.3).

EXHIBIT 4.2 The Relationship of Organization Design to Efficiency versus Learning Outcomes



COUNTERPOINT 4.3

Note that the division of labour is associated with control in *the name of efficiency* and not necessarily with the delivery of enhanced efficiency or effectiveness. At the extreme, the autocratic CEO may exert control through the micro-management of activities so that no significant decision is made without his or her agreement. Such control will likely produce a backlog of decision-making and delayed decisions that are based upon a distant and limited understanding of the specific circumstances. What is true of the autocratic CEO is also true, to a degree, of all organizations with a structure where those 'at the top' become remote from what is happening 'at the bottom'. The problems are compounded when senior management teams either assume that they are omniscient or rely upon managers who live in fear of providing unwelcome information.

Some of the most successful 'new economy' companies such as Google operate on a highly decentralized model, giving employees considerable discretion to collaborate and innovate.¹² At the same time, even in a decentralized company, a basic hierarchical structure usually remains in place even though it may be overlaid and obscured by an emphasis upon horizontal coordination. At the very least, shareholders require assurance that their investment is 'under control' as well as being deployed as effectively as possible by ensuring agility and responsiveness to emerging opportunities and challenges, as discussed in the Oracle case below. Increasingly, information technologies are used to monitor performance, ideally in real time, thereby facilitating centralized as well as decentralized communication to keep control of the diverse operations of global corporations. For example, Mexico's Cemex, discussed in Chapter 2, has installed computers with broadband internet connections into employees' homes, all linked to the company's intranet portal.¹³



ONLINE
COUNTERPOINT 4.2

Managers are, in principle, always searching for the best combination of vertical control and horizontal collaboration, centralization and decentralization, taking advantage of communications advances to perfect interaction both vertically and horizontally.

Vertical Information Linkages

Linkage is defined as the extent of communication and coordination between organizational elements. **Vertical linkages** are used to coordinate activities between the top and bottom of an organization and are designed primarily for control of the organization. Employees at lower levels are expected to carry out activities consistent with top-level goals, and top executives must be informed of activities and accomplishments at the lower levels. A variety of structural devices may be used to achieve vertical linkage, including hierarchical referral, rules, plans and formal management information systems.¹⁴

Hierarchical Referral The first vertical device is the hierarchy, or chain of command, which is illustrated by the vertical lines in Exhibit 4.1. If a problem arises that employees don't know how to solve, it can be referred up to the next level in the hierarchy. When the problem is solved, the answer is passed back down to lower levels. The lines of the organization chart act as communication channels.

Rules and Plans The next linkage device is the use of rules and plans. To the extent that problems and decisions are repetitious, a rule or procedure can be established so employees know how to respond without communicating directly with their manager. Rules provide a standard information source enabling employees to be coordinated without actually communicating about every task. A plan also provides standing information for employees. The most widely used plan is the budget. With carefully designed budget plans, employees at lower levels can be left on their own to perform activities within their resource allotment.

Vertical Information Systems A **vertical information system** is another strategy for increasing vertical information capacity. Vertical information systems include the periodic reports, written information and computer-based communications distributed to managers. Information systems make communication up and down the hierarchy more efficient. Vertical information systems are an important component of vertical control at software-maker Oracle.

In today's world of corporate financial scandals and ethical concerns, many top managers are considering strengthening their organization's linkages for vertical information and control. The other major issue in organizing is to provide adequate horizontal linkages for coordination and collaboration.

Horizontal Information Linkages

Horizontal communication overcomes barriers between departments and provides opportunities for coordination between employees to achieve unity of effort and organizational objectives. **Horizontal linkage** refers to the amount of communication and coordination horizontally across organizational departments. Its importance is articulated by comments made by Lee Iacocca when he took over the ailing Chrysler Corporation in the 1980s:

What I found at Chrysler were 35 vice presidents, each with his own turf ... I couldn't believe, for example, that the guy running engineering departments wasn't in constant touch with his counterpart in manufacturing. But that's how it was. Everybody worked independently. I took one look at that system and I almost threw up. That's when I knew I was in really deep trouble ... Nobody at Chrysler seemed to understand that interaction among the different functions in a company is absolutely critical. People in engineering and manufacturing almost have to be sleeping together. These guys weren't even flirting!¹⁵

During his tenure at Chrysler (from 1997 to 2007, DaimlerChrysler¹⁶), Iacocca pushed horizontal coordination to a high level. Everyone working on a specific vehicle project – designers, engineers and manufacturers, as well as representatives from marketing, finance, purchasing and even outside suppliers – worked together on a single floor so they could constantly communicate.

Horizontal linkage mechanisms often are not drawn on the organization chart, but they invariably form some element of organization structure. The following devices are structural alternatives that can improve horizontal coordination and information flow.¹⁷ Each device enables people to exchange information.



ONLINE BRIEF 4.2

IN PRACTICE

Oracle Corporation

In an era of decentralization and empowerment, Larry Ellison, co-founder of Oracle and CEO for over a quarter-century before he ‘partially’ stepped down in 2014 to become executive chairman and chief technical officer, doesn’t hesitate to proclaim his belief in stronger vertical control. Oracle got in trouble some years ago because sales managers around the globe were cutting backroom deals or hammering out private, individualized compensation agreements with salespeople in different countries. Today, all the terms, including sales contracts and commissions, are dictated from the top and spelled out in a global database. All deals must be entered into the database so they can easily be tracked by top managers. This is in accordance with today’s HR analytics.

The internet plays a key part in Oracle’s vertical information and control systems, by offering the power to centralize complex operations while also rapidly disseminating information all over the world. Oracle uses its own suite of internet software applications that work together on a global basis. All employees do their work via the internet, enabling the top management to carefully track, analyze and control the behaviour of each unit, manager and employee. Although many managers weren’t happy with the stronger top-down control, Ellison believed it was necessary to effectively manage a sprawling global corporation that was beginning to behave more like a bunch of separate companies. In addition, the system helps to circulate and ensure implementation

of standard rules and procedures across divisions. According to Chief Marketing Officer, Mark Jarvis, this ultimately provides for greater freedom for lower levels and prevents the hierarchy from becoming overloaded. ‘Once we have a standard set of global business practices’, Jarvis notes, ‘the [managers] can be allowed more scope for decision-making within the broad framework’.

Oracle’s 2004 acquisition of PeopleSoft, as well as other, smaller acquisitions, has increased the complexity of the organization, but Oracle’s top managers are focused on smooth integration through the use of vertical information systems. The company is working on developing a super-suite of software applications that combines the best features of products from Oracle, PeopleSoft and J.D. Edwards, and will allow for standardization and centralization across the enterprise. Oracle hopes that the new super-suite, dubbed Project Fusion, released in 2011 after six years of development, will allow its customers to automate an entire global infrastructure so that everything is linked and compatible and managers can get the information they need to effectively control the organization.¹⁸ The lengthy delays encountered by Project Fusion underline the difficulties inherent in complete vertical integration.¹⁹ Ellison has been criticized for making too many acquisitions, particularly troubled high-end computer hardware manufacturer Sun Microsystems in 2009. However, the company remains one of the Silicon Valley giants and has prospered with its emphasis on cloud-based computing. Revenues and profits have been fairly flat in the past several years, but at nearly \$40 billion in revenues and \$10 billion in net income in 2019, the company is well placed to continue responding successfully to industry transformation. Despite his announced semi-retirement, Ellison still speaks on behalf of the company, and his over \$71 billion in personal wealth ranked him in 2019 the seventh wealthiest person in the world.²⁰

Information Systems A significant method of providing horizontal linkage in today's organizations is the use of cross-functional information systems. Computerized information systems can enable managers or frontline workers throughout the organization to routinely exchange information about problems, opportunities, activities or decisions. For example, Siemens uses an organization-wide information system that enables 450,000 employees around the world to share knowledge and collaborate on projects to provide better solutions to customers. The information and communications division recently collaborated with the medical division to develop new products for the healthcare market.²¹

Employees may also be encouraged to use the company's information systems in order to build relationships across the organization, aiming to support and enhance ongoing horizontal coordination across projects and geographical boundaries. In 2007, The Samaritans, a UK charity which runs telephone helplines and other services for people in emotional distress, adopted a specially tailored Customer Management System (CRM) that consolidates information about its 17,000 volunteers and its financial supporters, as well as business partners. The system permits the organization's staff to work more efficiently, making linkages between different 'customers' of the organization in a way that was not possible before. Thus, for example, its business partners could be asked to launch an appeal for volunteers within their companies. In an indication of the increasing sophistication and professionalization of charitable organizational design, in 2013 Samaritans adopted a comprehensive CRM platform, thankQ, to manage its customer relationships.^{22, 23}

Direct Contact A higher level of horizontal linkage is direct contact between managers or employees affected by a problem. One way to promote direct contact is to create a special **liaison role**. A liaison person is located in one department but has the responsibility for communicating and achieving coordination with another department. Liaison roles often exist between engineering and manufacturing departments because engineering has to develop and test products to fit the limitations of manufacturing facilities. At Johnson & Johnson, CEO William C. Weldon set up a committee made up of managers from research and development (R&D) and sales and marketing. The direct contact between managers in these two departments enables the company to set priorities for which new drugs to pursue and market. Weldon also created a new position to oversee R&D, with an express charge to increase coordination with sales and marketing executives.²⁴ Another approach is to locate people close together so they will have direct contact on a regular basis.

Task Forces Liaison roles usually link only two departments. When linkage involves several departments, a more complex device such as a task force is required. A **task force** is a temporary committee composed of representatives from each organizational unit affected by a problem.²⁵ Each member represents the interests of a department or division and can carry information from the meeting back to that department.

Task forces can be an effective horizontal linkage device for temporary issues. They endeavour to solve problems by direct horizontal coordination and reduce the information load on the vertical hierarchy. Typically, they are disbanded after their tasks are accomplished or superseded.

Task forces have been used for everything from organizing the annual company picnic to solving expensive and complex manufacturing problems. Philips Corporation, discussed in Chapter 1, made use of multiple taskforces in its Operation Centurion to help restructure the organization as it began to face up to the need to respond to global challenges in the 1990s. The task force approach set Philips on a very long road to recovery, but in other cases, task forces have been used to give impetus to fundamentally flawed strategies. One example of a negative outcome of the task force approach was the Executive Automotive Committee formed by DaimlerChrysler CEO Jürgen Schrempp. This task force was set up specifically to identify ideas for increasing cooperation and component sharing among Mercedes, Chrysler and Mitsubishi (in which DaimlerChrysler owned a 37 per cent stake). The task force started with a product road map, showing all Mercedes, Chrysler, Dodge, Jeep and Mitsubishi vehicles to be launched over a ten-year period, along with an analysis of the components they would use, so task force members could identify overlap and find ways to

share parts and cut time and costs.²⁶ Unfortunately, Mitsubishi was in serious trouble, both because of the languishing Japanese economy and a number of strategic errors. As Mitsubishi sank further into crisis, the logic for production integration with the company was undermined, and eventually the company was bailed out, resulting in the loss of most of DaimlerChrysler's investment, and contributing to Schrempp's departure from DaimlerChrysler.²⁷

Full-time Integrator A stronger horizontal linkage device is to create a full-time position or department solely for the purpose of coordination. A full-time **integrator** frequently has a title, such as product manager, project manager, programme manager or brand manager. Unlike the liaison person described earlier, the integrator does not report to one of the functional departments being coordinated. He or she is located outside the departments and has the responsibility for coordinating several departments, or even, in the case below, in integrating internal departments, suppliers and customers.

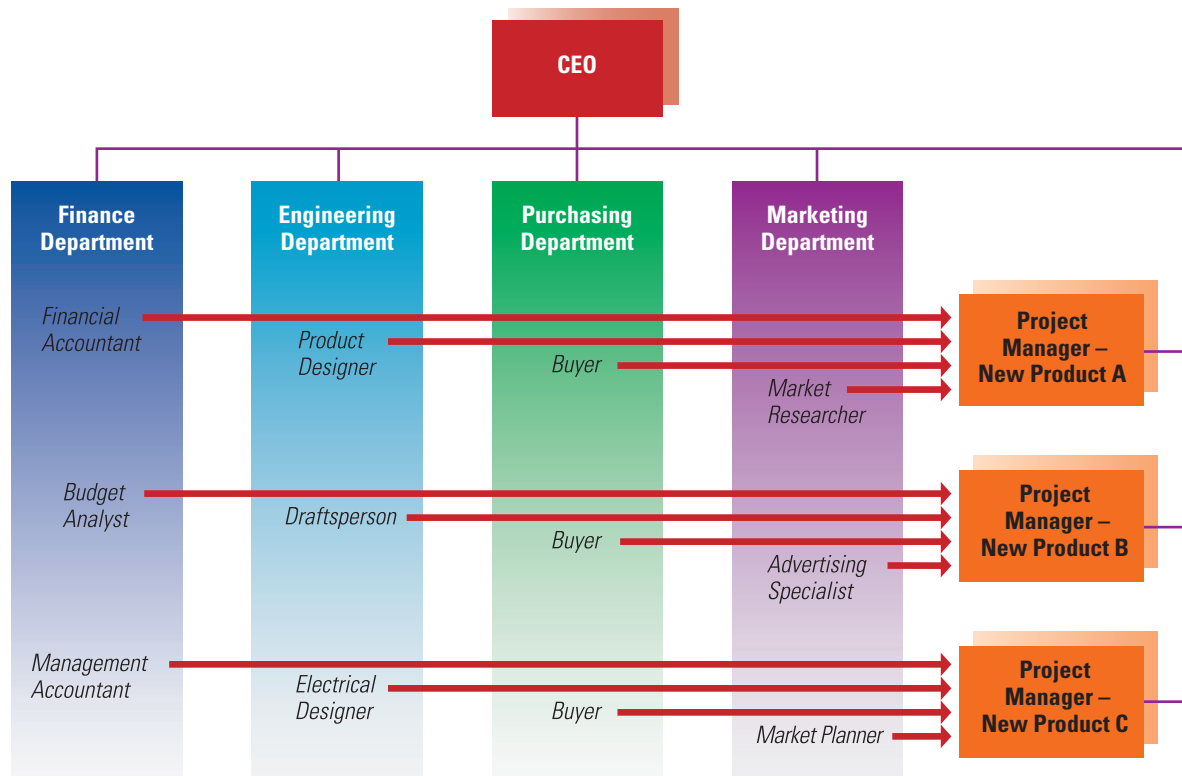
Kalmar Industries is the world's leading supplier of cargo handling equipment to ports, terminals and intermodal facilities, and is a subsidiary of the Finland-headquartered, globally present Cargotec Corporation. Kalmar provides customized cargo solutions to large-scale customers, with 2014 sales of about €1.5 billion. As in many modern manufacturing enterprises, most of Kalmar's production is outsourced, with the company heavily involved in system integration, liaising closely both with suppliers and customers to ensure delivery and support of the customized product.²⁸ Kalmar's managers realized that although the company had strong relationships with its suppliers and customers, there were many inefficiencies due both to communications misunderstandings and logistical costs of manufacturing items in separate locations and transporting them to Kalmar for assembly. In order to respond to this issue, Kalmar established Product Supply Centres (PSCs), where key suppliers co-located their production facilities for specific Kalmar products. This approach allowed Kalmar to respond much more quickly and effectively to customers' specific requests.²⁹

An integrator or integrating department can be responsible for an innovation or change project, such as coordinating the design, financing and marketing of a new product. An organization chart that illustrates the location of project managers for new product development is shown in Exhibit 4.3. The project managers are drawn to the side to indicate their separation from other departments. The arrows indicate project members assigned to the new product development. New Product A, for example, has a financial accountant assigned to keep track of costs and budgets. The engineering member provides design advice, and purchasing and manufacturing members represent their areas. The project manager is responsible for the entire project. He or she sees that the new product is completed on time, is introduced to the market and achieves other project goals. The horizontal lines in Exhibit 4.3 indicate that project managers do not have formal authority over team members with respect to giving pay raises, hiring or firing. Formal authority rests with the managers of the functional departments.

Integrators in most companies have a lot of responsibility but little authority. The integrator has to use expertise and persuasion to achieve coordination. He or she spans the boundary between departments and must be able to get people together, maintain their trust, confront problems, and resolve conflicts and disputes in the interests of the organization.³⁰

Teams Project teams tend to be the strongest horizontal linkage mechanism. **Teams** are permanent task forces and are often used in conjunction with a full-time integrator. When activities between departments require strong coordination over a long period of time, a cross-functional team is often the solution. Special project teams may be used when organizations have a large-scale project, a major innovation or a new product line.

Imagination Ltd, Britain's largest design firm, relies heavily upon teamwork. At the beginning of each project, Imagination puts together a team of designers, writers, artists, marketing experts, information specialists and representatives of other functional areas to carry out the entire project from beginning to end. Hewlett-Packard's Medical Products Group uses *virtual cross-functional teams*, made up of members from various countries, to develop and market medical products and

EXHIBIT 4.3 Project Manager Location in the Structure

services such as electrocardiograph systems, ultrasound imaging technologies and patient monitoring systems.³¹ A **virtual team** is one that is made up of organizationally or geographically dispersed members who are linked primarily through advanced information and communications technologies. Members frequently use online collaborative technologies to work together, rather than meeting face to face.³²

**ONLINE BRIEF 4.3**

Volvo, the Sweden-headquartered truck manufacturer, has an enviable reputation for innovative management practices and particularly with fostering strong teamwork. The Dutch researchers Ben Kuipers and Marco de Witte took a close look at teamwork at Volvo's cab manufacturing plant in Umeå in northern Sweden to better understand how successful teamwork develops, what they call in *Volvoment*. They identified four dimensions to teamwork:

- Job enlargement: encouraging employee multifunctionality and thus the elimination of inefficiency within the team
- Job enrichment: the redesign of the supervision system to delegate management responsibilities to the team members
- Cooperation: members work together as a team, building on communication and a sense of shared responsibility
- High performance: the team is able to work together to resolve non-routine issues, requiring collaborative innovation.

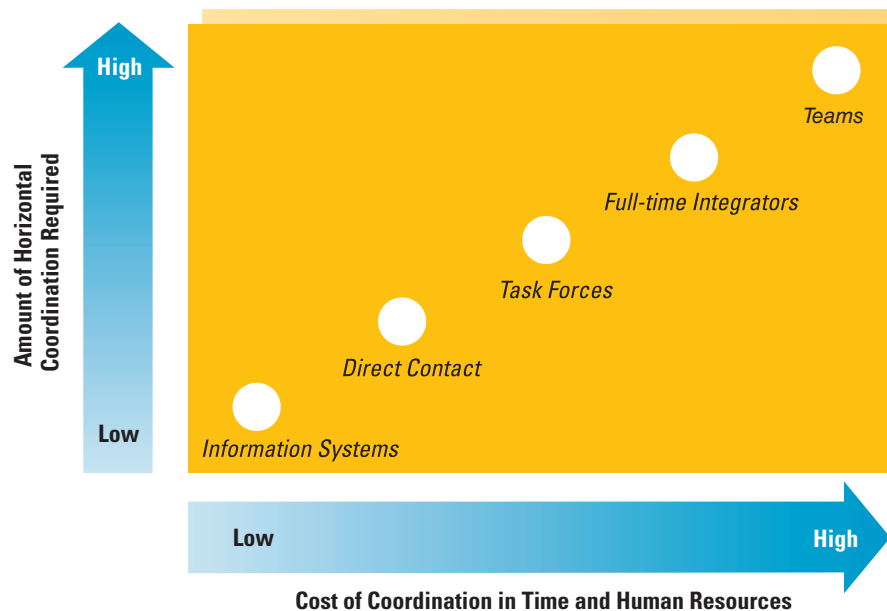
Unlike some other studies of teamwork, Kuipers and de Witte found that these four aspects of teamwork don't occur sequentially, one after the other, but through an interconnected process. They showed that the higher the level of teamwork achieved, the better quality of product, higher productivity and employee satisfaction. Truly a win-win.³³

Teambuilding has become so important to companies that it has spawned a whole new business opportunity in teamwork development. Many organizations hire expert companies to support the teambuilding process. Often, the teambuilding includes some physically challenging activities that require close collaboration to achieve success, such as sailing a boat together, overcoming an outdoor obstacle course or even performing a daredevil circus act. Airbus, the European aeroplane manufacturer, has management staff working in several different countries who rarely meet face to face, so a building team spirit is a challenge, even with rapid advances in web-based communications. At a recent annual away day, the company hired a teambuilding company, Blue Hat, which organized a 'Da Vinci Code Challenge' in which small teams had to work together to find the famous chalice and elixir of life.³⁴

Exhibit 4.4 summarizes the mechanisms for achieving horizontal linkages. These devices represent alternatives that managers can select to increase horizontal coordination in any organization. The higher-level devices provide more horizontal information capacity, although the cost to the organization in terms of time and human resources is greater. If horizontal communication is insufficient, departments will find themselves out of synchronization, resulting in inefficiency and ineffectiveness.



EXHIBIT 4.4 Ladder of Mechanisms for Horizontal Linkage and Coordination



Organization Design Alternatives

Overall, the design of organization structure indicates required work activities, reporting relationships and departmental groupings.

Required Work Activities

Departments are created to perform tasks considered important to the company. For example, in a manufacturing company, work activities are organized into a range of functions, such as a human resource department to recruit and train employees, a purchasing department to obtain supplies and raw materials, a production department to build products, a sales department to sell products, and so forth. As organizations grow larger, their organizational structure often becomes more complex as different functions are added. For example, before the 1980s, many companies had large divisions ('pools') of secretarial staff. Today, most employees do much of their

own administrative work so secretaries have, in effect, been replaced with information technology departments whose purpose is to deliver the smooth operation of the computers and networks used to create and transmit data within and beyond the company.

Reporting Relationships

Hierarchical reporting relationships, often called the *chain of command*, are represented by vertical lines on an organization chart. The chain of command is typically an unbroken line of authority that links all persons in an organization and shows who reports to whom. Although organizations have emphasized flattening of hierarchy in recent years, this typically reflects a reduction in the number of vertical levels rather than an absence of hierarchy.

Departmental Grouping Options

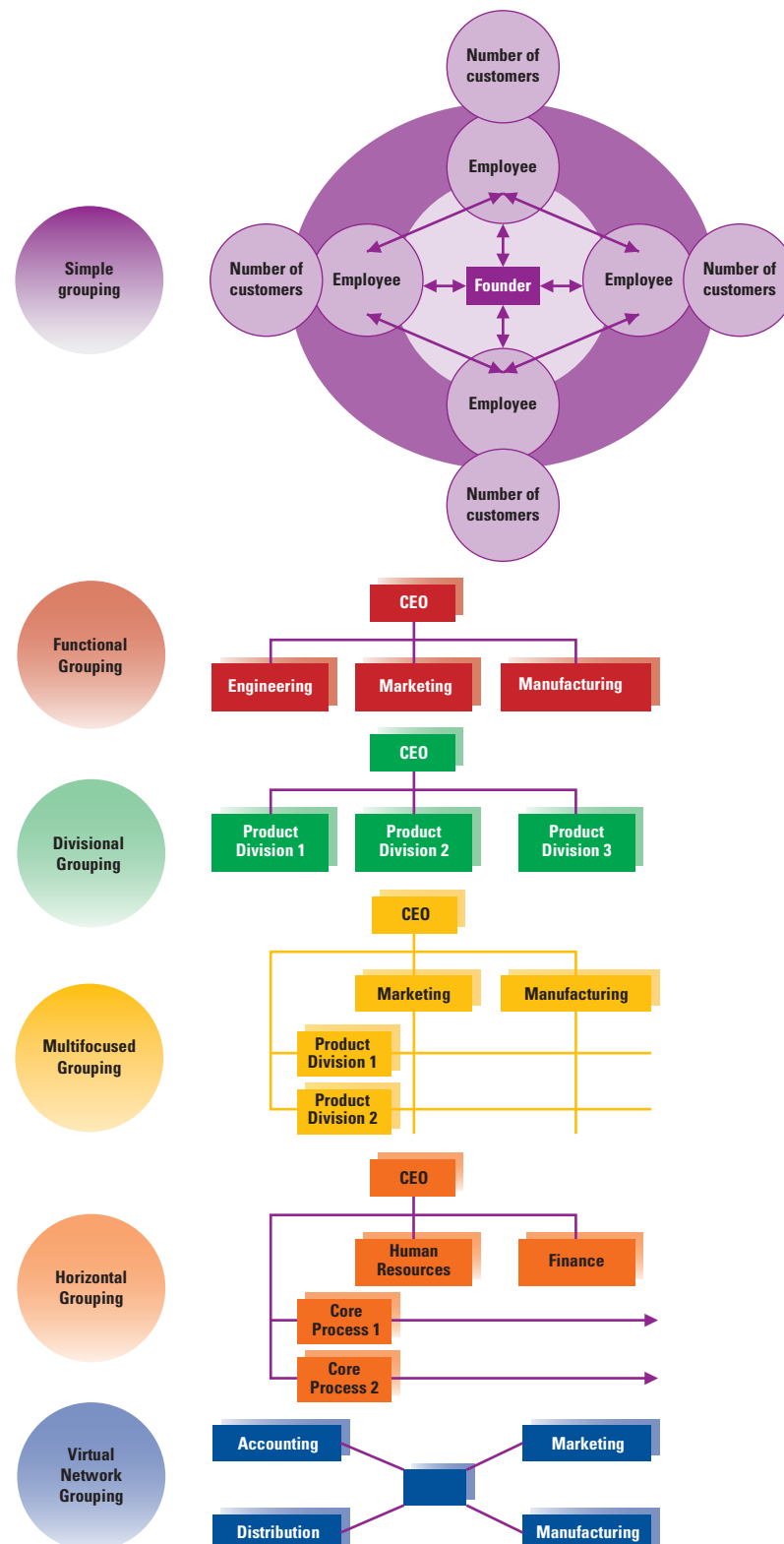
Early stage companies are typically loosely structured with new functions added (and removed) on an *ad hoc* basis (simple structure) as the organization finds its market niche and shapes itself around the needs of operating in that niche. As the organization grows and becomes more stable, departmental groupings are typically established to provide efficiency and predictability to operations. Some options for departmental grouping, including **simple grouping**, **functional grouping**, **divisional grouping**, **multifocused grouping**, **horizontal grouping** and **virtual network grouping**, are illustrated in Exhibit 4.5. **Departmental grouping** affects employees because they share a common supervisor and common resources, are jointly responsible for performance, and tend to identify and collaborate with one another.³⁵ For example, a credit manager's perspective might well change if she moves from the finance department to the marketing department. As a member of the marketing department, the credit manager would work more closely with salespeople, understanding the importance of increasing sales and thus becoming more willing to see credit as a sales tool than when she was located in the finance department.

Simple grouping means people work together without any formal reporting relationship. The structure is often used by founders and any other entrepreneurial persons. The organization is very small with only a few employees. The founder is the main person of the organization and the few employees work together informally. The employees make sure that the organization can adapt to customers as well as any other changes in the environment very quickly.

Functional grouping places together employees who perform similar functions or work processes or who bring similar knowledge and skills to bear. For example, all marketing people work together under the same supervisor, as do manufacturing and engineering people. All people associated with the assembly process for generators are grouped together in one department. All chemists may be grouped in a department different from biologists because they represent different disciplines.

Divisional grouping means people are organized according to what the organization produces. All people required to produce toothpaste – including personnel in marketing, manufacturing and sales – are grouped together under one executive. In many large conglomerates, some product or service lines have entirely separate identities from that of the parent company. An example, is that of the French-headquartered Accor group of hotels.

Multifocused grouping means an organization embraces two structural grouping alternatives simultaneously. These structural forms are often called matrix or *hybrid*. They will be discussed in more detail later in this chapter. An organization may need to group by function and product division simultaneously or perhaps by product division and geography.

EXHIBIT 4.5 Structural Design Options for Grouping Employees into Departments

Source: Adapted from David Nadler and Michael Tushman, *Strategic Organization Design* (Glenview, IL.: Scott Foresman, 1988), 68.

Horizontal grouping means employees are organized around core work processes, the end-to-end work, information and material flows that provide value directly to customers or support strategic development. All the people who work on a process are brought together in a group rather than being separated into functional departments. Frequently, new product development is organized through horizontal and/or cross-functional groups, ensuring for example that product design takes into account manufacturing challenges. India's first sport utility vehicle, the Mahindra Scorpio, was developed through the work of a cross-functional team of 120 staff from across the company.³⁶ Horizontal groupings are often implemented in addition to traditional vertical working relationships. In the West African state of Senegal, for example, representatives of the various government agencies involved in agricultural sector management form part of an 'inter-professional committee' that ensures coordinated agricultural policy and programming.³⁷ Horizontal grouping may be adopted in civil society organizations to fit with the egalitarian ethos of the members. For example, the Argentinian Movimiento de Trabajadores Desocupados Solano (MTD Solano) is a network of unemployed people dedicated to improving the circumstances of the unemployed and other people in Argentina. In line with its members' belief that social inequality is the cause of their problems, MTD Solano rejects a leadership hierarchy and operates through horizontal groupings of different groups of social activists.³⁸

Virtual network grouping is one of the more recent approaches to departmental grouping. With this grouping, the organization is a loosely connected cluster of separate components. In essence, departments are separate organizations that are electronically connected for the sharing of information and completion of tasks. Departments can be spread all over the world rather than located together in one geographical location. In recent years there has been a rapid increase in the use of this type of structure, implemented through the re-engineering of business processes. Business process outsourcing (BPO) is an example where organizational functions such as customer service can actually be placed outside the firm, and even in another part of the world. Virtual network organizations are one example of what some organization thinkers, such as David Boje, describe as 'post-modern' organizations: organizations that are fluid, flexible and infinitely reconfigurable according to the needs of a rapidly changing environment. In today's rapidly evolving environment, it is not uncommon for organizations to completely change their identity and business focus over a period of only a few years. IBM for example has shifted its core business away from computer manufacture towards consulting over the past 20 years.³⁹

The organizational forms described in Exhibit 4.5 provide a range of overall options within which the organization chart is drawn and the detailed structure is designed, although as always, various *hybrid* and *anomalous* grouping approaches are found in real life. Each structural design alternative has significant strengths and weaknesses, to which we now turn.

Simple, Functional, Divisional and Geographical Designs

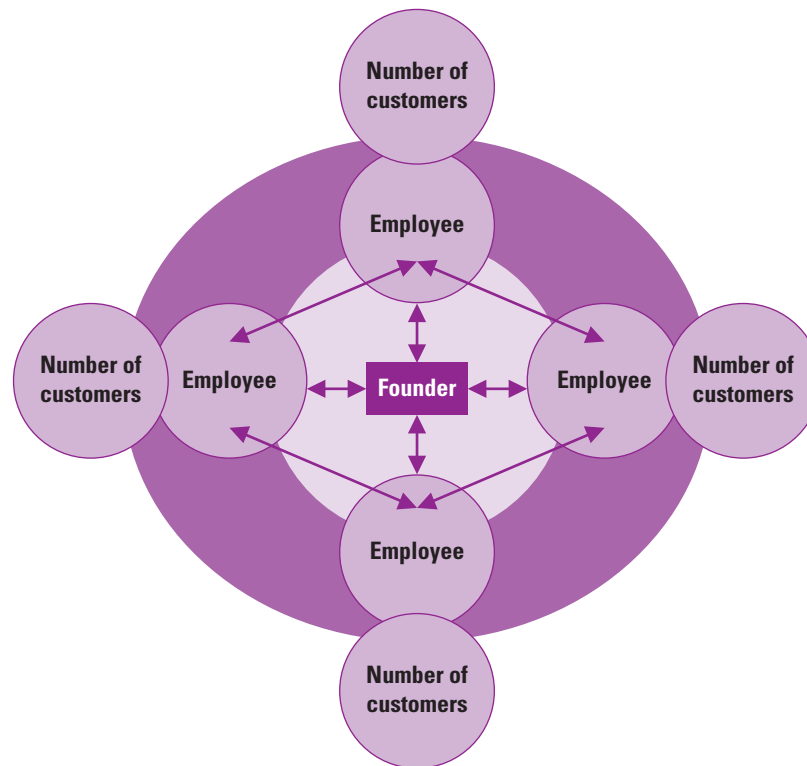
Functional grouping and divisional grouping are the two most common approaches to structural design.

Simple Structure

This is often the first kind of structure used by a founder when a new organization is formed. The organization consists of the founder and a small number of employees. In this type of structure there are no formal hierarchical reporting relationships, therefore there is no chain of command. The founder tends to devote his or her full energies to the technical activities of production and marketing and manages the simple structure informally. The running of the organization is therefore based on the personal values of the founder rather than on bureaucratic rules.

The organization creates growth by being creative when it comes to new products or services. Generally, employees learn from their close contact with individual customers. Each employee handles their own customers, and because there is no formal division of labour, each employee carries out many different tasks; that is, the primary customer contact, sales, marketing developments and after sales. Consequently, the organization becomes very good at differentiating products but not so good at scaling production. This is illustrated in Exhibit 4.6.

EXHIBIT 4.6 The Simple Structure



A special characteristic of the simple organization is its specific focus on each customer together with the close contact between employees and their customers. As a result, employees often develop products individually for each customer leading to a large product portfolio.

The simple organization is driven forward by the enthusiasm of the founder and employees—both are committed to the business and usually work many hours without extra pay. Therefore, the coordinating mechanism is good friendship and adaption to the demands of customers.

The shortcomings of the simple structure are primarily two fold. The first is the importance of the founder. Because the new entity is the brain child of the founder, he or she is part of everything going on. Whilst this works well initially, as the organization grows, it becomes harder for the founder to be involved with everything and retain a complete overview.

The second is the absence of formal reporting relationships. This is not a problem in the beginning as employees are enthusiastic, committed and happy to work without rules or procedures. However, when the organization is successful and grows, formal reporting relationships as well as procedures become more important to the effective running of the business. This can be an issue because the founder may be more interested in creating new products and fulfilling customer demands than obtaining scale economies.

The different strengths and weaknesses of the simple structure are illustrated in Exhibit 4.7.

EXHIBIT 4.7 Strengths and Weaknesses of the Simple Structure

Strengths	Weaknesses
1. Flexibility and quick response to changes in customer demands 2. Enables employees to focus on product development (delivering value to customers) 3. High employee commitment, loyalty and adaptability 4. High responsiveness with commitment to end result 5. Strong clan and adaptive culture 6. Good for small companies	1. Hard to know which employees do what 2. Limited use of cross-abilities and development 3. Eliminates economies of scale among employees 4. Big product portfolio 5. Sometimes difficult to coordinate across different employees and customers

An example of a successful simple organization is the Danish company Martin Professional, which is a manufacturer and distributor of stage and architectural lighting and effects fixtures. Entrepreneur Peter Johansen discovered in 1978 that he could produce a smoke generator from a coffee maker. By 1986 he had founded the company Martin Professional in Aarhus, Denmark, and begun producing fog machines; he added a small selection of disco lights the next year. Originally using the basement of his house, Johansen moved the company into a stockroom, at which point four or five more employees were hired to meet growing demand. The company continued to be successful and to grow, but due to its simple structure it ran into some of the shortcomings mentioned above.

Functional Structure

In a **functional structure**, activities are grouped together by common function from the bottom to the top of the organization. All engineers are located in the engineering department, and the vice president of engineering is responsible for all engineering activities. The same is true in marketing, R&D and manufacturing. An example of the functional organization structure was shown in Exhibit 4.1 earlier in this chapter.

With a functional structure, all human knowledge and skills with respect to specific activities are consolidated, providing a valuable depth of knowledge for the organization. This structure is most effective when in-depth expertise is critical to meeting organizational goals, when the organization needs to be controlled and coordinated through the vertical hierarchy, and when efficiency is important. The structure can be quite effective, especially if there is little need for horizontal coordination. Exhibit 4.8 summarizes the strengths and weaknesses of the functional structure.

The functional structure can improve economies of scale by concentrating specialists in groups in a common location and by sharing facilities. Producing all products in a single plant, for example, may strengthen a financial case for the plant to acquire the latest machinery. Constructing only one facility instead of separate facilities for each product line can reduce duplication and waste. The functional structure may also promote in-depth skill development of employees. Employees are exposed to a range of functional activities within their own department.⁴⁰

The potential shortcoming of the functional structure is a slow response to environmental changes that require coordination across departments. The vertical hierarchy can become

isolated or overloaded. Decisions pile up as top managers lack effective means of coordination across specialist functions. A functional structure may also impede innovation when this requires coordination.

Many small- and medium-sized organizations in which the focus is upon core competencies, operate with a functional structure. Consider the *In Practice* case of Real Hotel Company Ltd, a medium-sized UK-based hotel chain, active in Britain and Europe. Real Hotels had made losses for several years, and in 2006 its top management decided to reorganize into a clear functional structure as one of a series of steps designed to bring the company back into the black.

EXHIBIT 4.8 Strengths and Weaknesses of Functional Organization Structure

Strengths	Weaknesses
1. Allows economies of scale within functional departments 2. Enables in-depth knowledge and skill development 3. Enables organization to accomplish functional goals 4. Is best with only one or a few products	1. Slow response time to environmental changes 2. May cause decisions to pile on top, hierarchy overload 3. Leads to poor horizontal coordination among departments 4. Results in less innovation 5. Involves restricted view of organizational goals

Source: Adapted from Robert Duncan, 'What Is the Right Organization Structure? Decision Tree Analysis Provides the Answer', *Organizational Dynamics* (Winter 1979), 429.

Functional Structure with Horizontal Linkages

During the late years of the twentieth century, criticism of extended hierarchical structures promoted a shift towards flatter, more horizontal structures, as business activities became more complex and companies identified the potential for restructuring business processes to achieve synergies. Organizations may compensate for the vertical functional hierarchy by installing horizontal linkages, as described earlier in this chapter. Managers may improve horizontal coordination by using information systems, direct contact between departments, full-time integrators or project managers, task forces or teams (illustrated in Exhibit 4.3). One interesting use of horizontal linkages occurred at Karolinska Hospital in Stockholm, Sweden, which once had 47 functional departments. Even after top executives cut that down to 11, coordination was still inadequate. The team set about reorganizing workflow at the hospital around patient care. Instead of bouncing a patient from department to department, Karolinska now envisions the illness to recovery period as a process with 'pit stops' in admissions, X-ray surgery and so forth. The most interesting aspect of the approach is the new position of nurse coordinator. Nurse coordinators serve as full-time integrators, troubleshooting transitions within or between departments. The improved horizontal coordination dramatically improved productivity and patient care at Karolinska.⁴¹ Karolinska is effectively using horizontal linkages to overcome some of the disadvantages of the functional structure. In 2004, Karolinska merged with another large hospital. Some progress has been made in effectively integrating the hospitals, particularly in the area of IT, but integration problems remain.



ONLINE BRIEF 4.4



IN PRACTICE

The Real Hotel Company PLC

When Michael Praeger took over as CEO of the Real Hotel Company in early 2007, he knew he had a difficult task on his hands. His previous experience with several international chains would stand him in good stead on the hotel side of the business, but the company needed radical action to halt years of poor financial performance and failed restructuring efforts.

The Real Hotel Company is a British-headquartered company with its main business operating budget to mid-range hotels. Since being founded as Friendly Hotels in the mid-1980s by the iconic hotelier Henry Edwards, in the late 1990s the company secured the European master franchise for the global chain of budget hotels, Choice Hotels International, meaning the company was operating over 300 hotels across the continent. The business almost immediately went into the red and Henry Edwards left the company he had founded. Despite a dizzying series of name changes, capital raising efforts, hotel refurbishments, attempts to sell out and even a near

bankruptcy in 2001, the business never produced satisfactory results. By the time Michael Praeger arrived, the company had already decided to exit the franchise agreement with Choice Hotels, which it successfully did in late 2006. But with the great majority of company-operated hotels now gone, the hybrid functional-geographic management structure seemed top-heavy. Furthermore, the company now needed to focus on building its own brands, which had been a lower priority while the company was trying to manage all the Choice Hotel franchises.

Praeger slimmed down the management structure, eliminating the geographic divisions in favour of a functional structure. He gave up development of Sleep Inn, a brand that had never really caught on, and focused most of the company's efforts on the new Purple Hotel label – a contemporary brand that would appeal to a younger clientele interested in value, but also style. Real Hotels invested heavily in its sales and marketing functions, and pondered exiting completely from the traditional budget sector. The company now had a manageable portfolio of properties and a simple functional management structure.

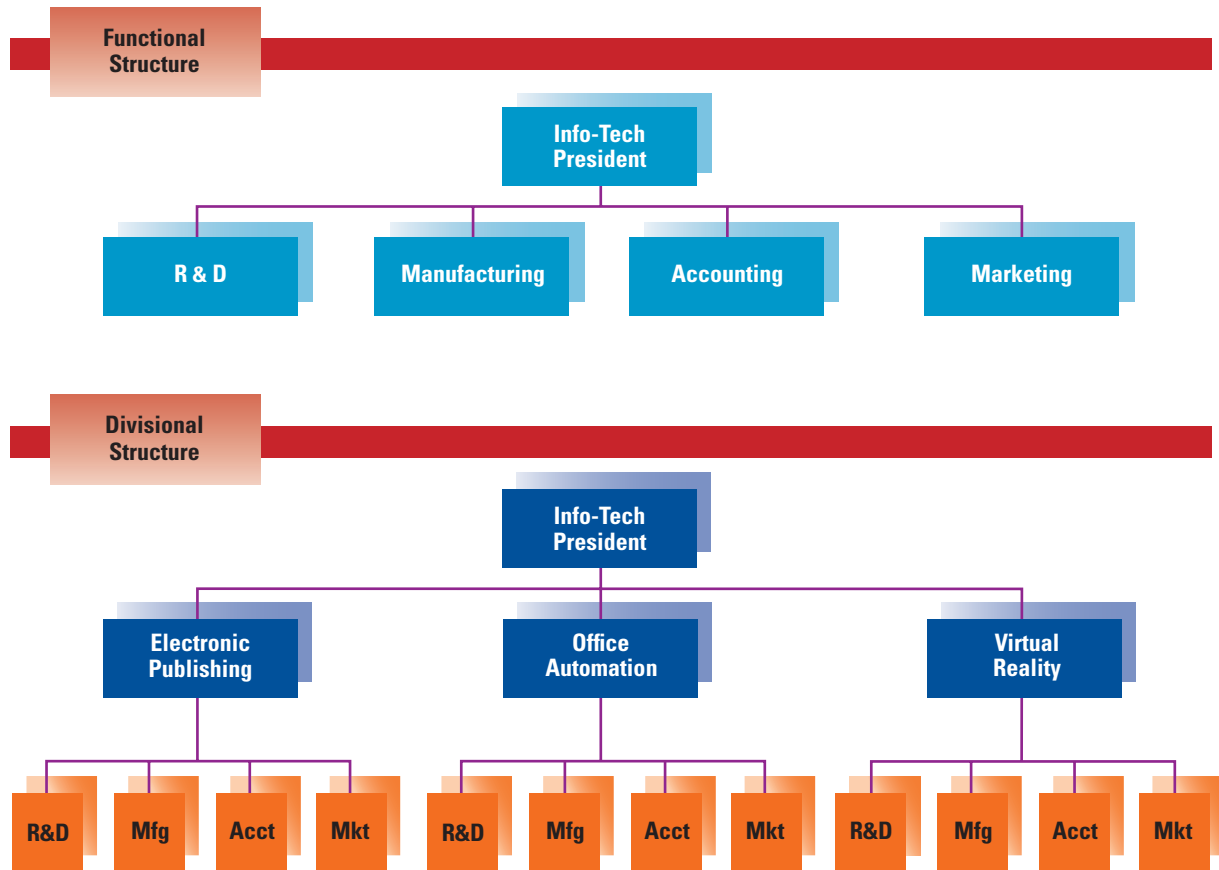
In early 2008, revenues started to pick up. However, with the onset of a serious recession during 2008, all the efforts proved too little, too late, and in January 2009 Real Hotels was forced to call in the administrators. The chain was carved up, with big competitors like Travelodge picking up the most profitable properties. Was Praeger's strategy flawed, or did the global economic situation mean that the turnaround never had much chance of success?⁴²

In today's rapidly changing organizational environment, managers are constantly challenged to adapt and redesign organizational structures and systems to meet new operational challenges.⁴³ A more recent initiative involved removing 500 beds from the hospital and outsourcing them to private companies who would operate them within community-based facilities. This privatization approach to public services is being attempted in many countries, but the jury is still out as to whether it really delivers better value for money; a 2011 independent study of Swedish privatization initiatives sponsored by a pro-business research group concluded that it did not deliver savings.⁴⁴

Divisional Structure

The term **divisional structure** is used here as the generic term for what is sometimes called a *product structure* or *strategic business units*. With this structure, divisions can be organized according to individual products, services, product groups, major projects or programmes, divisions, businesses or profit centres. The distinctive feature of a divisional structure is that grouping is based on organizational outputs.

The difference between a divisional structure and a functional structure is illustrated in Exhibit 4.9. The functional structure can be redesigned into separate product groups, and each

EXHIBIT 4.9 Reorganization from Functional Structure to Divisional Structure at Info-Tech

group contains the functional departments of R&D, manufacturing, accounting and marketing. Coordination across functional departments within each product group is maximized. The divisional structure promotes flexibility and change because each unit is smaller and can adapt more flexibly to opportunities and threats. Moreover, the divisional structure *decentralizes* decision-making as the lines of authority converge at a lower level in the hierarchy. The functional structure, by contrast, is *centralized* as it forces decisions all the way to the top before a problem affecting several functions can be resolved.

Strengths and weaknesses of the divisional structure are summarized in Exhibit 4.10. The divisional organization structure is excellent for achieving coordination across functional departments. It works well when organizations can no longer be adequately controlled through the traditional vertical hierarchy and when goals are oriented towards adaptation and change. Giant, complex organizations such as Nestlé, General Electric and India's Tata Corporation are subdivided into a series of smaller, self-contained organizations for better control and coordination. In these large companies, the units are sometimes called divisions, businesses or strategic business units. The structure at Johnson & Johnson includes 204 separate operating units, including McNeil Consumer Products, makers of Tylenol; Ortho Pharmaceuticals, which makes Retin-A and contraceptive pills; and J & J Consumer Products, the company that manufactures globally known brands such as Johnson's Baby Shampoo and Band-Aids. Each unit is a separately chartered, autonomous company, although operating under the guidance of Johnson & Johnson's corporate headquarters.⁴⁵

EXHIBIT 4.10 Strengths and Weaknesses of Divisional Organization Structure

Strengths	Weaknesses
1. Suited to fast change in unstable environment 2. Leads to customer satisfaction because product responsibility and contact points are clear 3. Involves high coordination across functions 4. Allows units to adapt to differences in products, regions, customers 5. Best in large organizations with several products 6. Decentralizes decision-making	1. Eliminates economies of scale in functional departments 2. Leads to poor coordination across product lines 3. Eliminates in-depth competence and technical specialization 4. Makes integration and standardization across product lines difficult

Source: Adapted from Robert Duncan, 'What Is the Right Organization Structure? Decision Tree Analysis Provides the Answer', *Organization Dynamics* (Winter 1979), 431.

Many government organizations use a divisional structure. The UK National Health Service has made extensive organizational changes in order to meet service improvement targets, such as the requirement that Emergency Department patients will receive treatment within a specified time. Harrap and Gillies describe how one hospital reorganized from many small departments into five large divisions, each equipped with clear criteria for success and headed by a member of the senior management team.⁴⁶ The computer software company Microsoft uses a divisional structure to develop and market its different products.

The divisional structure has several strengths that are of benefit to Microsoft.⁴⁷ This structure is in principle responsive to fast changing, unstable conditions and can provide high product or service visibility. Each product line 'belongs' to a specific division, so customers can more easily identify and reach the right division to resolve any issues they have. In principle, each product can adapt to the requirements of individual customers or regions. The divisional structure is typically adopted in organizations that have multiple products or services and enough personnel to staff separate functional units. At firms like the Dutch-headquartered Philips Electronics (see *A Look Inside* in Chapter 1), the US-headquartered Johnson & Johnson pharmaceutical and personal hygiene company and Microsoft, decision-making is pushed down to the lowest levels. Each division is small enough to respond rapidly to changes in the market. Often, European firms favour greater decentralization to both product and geographic divisions than is the case with North American corporations.

One disadvantage of using divisional structuring is the loss of economies of scale. Instead of 50 research engineers sharing a common facility in a functional structure, ten engineers may be assigned to each of five product divisions. The critical mass required for in-depth research can be lost, and physical facilities have to be duplicated for each product line. Another problem is that product lines become separate from each other, and coordination across product lines can be difficult. An executive from Johnson & Johnson said, 'We have to keep reminding ourselves that we work for the same corporation'.⁴⁸ Excessive decentralization at Philips Electronics has been blamed for depriving the company of a coherent and comprehensive response to the onslaught of Japanese electronics goods from the 1970s onwards, with the result that Philips lost its dominant position in the European market.⁴⁹



IN PRACTICE

Microsoft

Bill Gates co-founded Microsoft in 1975 and built it into the most profitable technology company in the world. But as the company grew larger, the functional structure was just too slow and inflexible for a large organization operating in the fast-moving technology industry. Employees began complaining about the growing bureaucracy and snail's pace decision-making.

To speed things up and better respond to environmental changes, in 2002 top executives created seven business units (divisions), the Information Worker business, Microsoft Business Solutions (MBS), the Home and Entertainment Division, the Mobile and Embedded Devices Division, the Windows Client, Windows Server and the Tools group, built around the company's major products. Each division was run by a general manager and contained most of the functions of a stand-alone company, including product development, sales, marketing and finance. The division system was further streamlined three years later when the seven units were replaced by three core divisions. The Windows, MSN and Server groups became the *Microsoft Platform Products and Services Division*; the Information Worker and MBS groups were merged into the *Microsoft Business Division*; and the Mobile and Embedded Devices, and the Home and Entertainment division became the *Microsoft Entertainment and Devices Division*.⁵⁰

What really made the new structure revolutionary for Microsoft was that the division heads were given the freedom and authority to run the businesses and spend their budgets as they saw fit to meet goals. The general managers and chief financial officers for each division set their own budgets and manage their own profit and loss statements. Previously, the two top executives, Bill Gates and Steven Ballmer, were involved in practically every decision, large and small. Managers of the divisions are charged up by the new authority and responsibility. One manager said he feels 'like I am running my own little company'.

Nevertheless, Microsoft continues to lose ground to other players with very varied corporate structures, ranging from Apple, which was famously centralized in the hands of the late Steve Jobs, to Google, which operates a loose structure with substantial employee discretion. In particular, Microsoft has found it difficult to break away from its reliance on the Windows PC platform. Despite producing some attractive products in its Surface tablet and Windows Phone ranges it finds difficulty gaining traction in product areas where others have already built a reputation and market dominance.

In response, in 2013, Microsoft decided to emulate Apple, dissolving its divisions and creating what it called One Microsoft. The hope is that this approach will enable the company to leverage its competitive advantages across the different product categories by focusing much more on innovation rather than strong-arm sales tactics. Since this change of structure Microsoft has been able to grow its crucial cloud business and has released Windows 10.⁵¹

It is clear that, although corporate structure is important and needs to fit with the company's competitive needs, it is only one determinant of success.⁵²

From the early years of the twenty-first century, Microsoft began losing ground to Google. While Microsoft, with its divisional structure, had effectively countered competitive software, Google represented a new business model driven through search engine revenues. Without a coordinated approach, competitive threats can emerge from unexpected sources like this, and businesses may find themselves poorly placed even to identify the threat, let alone develop an effective response.

Companies such as Sony and Xerox have a large number of divisions and can have severe problems with horizontal coordination. It has been alleged that Sony lost out in the business of digital media products, despite its early advantage with the Walkman device, partly because of poor coordination. In the 2000s, Apple's iPod quickly captured 60 per cent of the US market versus only 10 per cent for Sony's Walkman range, a domination that Apple was then able to extend

into the mobile phone and tablet categories. The digital music business depends on seamless coordination. Sony's Walkman was unable to recognize some of the music sets made with early versions of the company's SonicStage software introduced in 2001. It wasn't until version 3.4 of SonicStage that compatibility was achieved – demonstrating a disconnection with the company's division selling music downloads. By 2007, Sony had decided to wind down its music download division, having lost the battle with Apple. In response to difficulties such as this, Sony moved decisively to more effectively integrate its disparate product divisions.⁵³

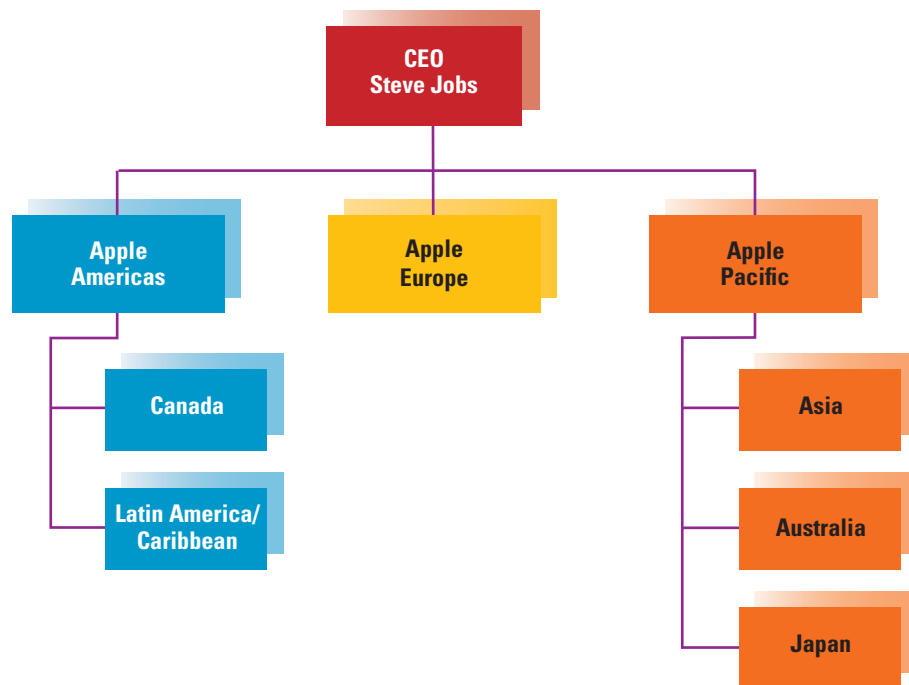
Unless effective horizontal mechanisms are in place, a divisional structure can be inefficient and unresponsive. One division may produce products or programmes that are incompatible with products sold by another division. Customers are frustrated when a sales representative from one division is unaware of developments in other divisions. Task forces and other linkage devices may be introduced to coordinate across divisions. A lack of technical specialization is also a problem in a divisional structure. Employees may identify with the product line rather than with a functional specialty. R&D personnel, for example, tend to be employed to undertake research to benefit a particular product line rather than basic research to benefit the entire organization.

Geographical Structure

Another basis for structural grouping is the organization's users or customers. The most common structure in this category is geography. Each region of the country may have distinct tastes and preferences. Each geographic unit includes all functions required to produce and market products or services in that region. This structure is particularly common in large NGOs and other civil society organizations that depend upon a close connection with volunteers, supporters or clients at the local level, such as the UK's Citizen's Advice Bureaux and Alcoholics Anonymous, and most political parties frequently use a type of geographical structure, with a central headquarters and semi-autonomous local units. The national or international headquarters organization provides brand recognition, coordinates fundraising services and handles some shared administrative functions, while day-to-day control and decision-making is decentralized to a greater or lesser extent to local or regional units.⁵⁴

Multinational corporations (MNCs) often create self-contained units for different countries and parts of the world. As MNCs integrate acquisitions into corporate structure, they often move from discrete functional units that were formerly stand alone businesses, towards geographic coordination. For example, in 2008, Kingfisher, the UK-headquartered multinational home renovation supplies retailer, best known for its B&Q stores, responded to disappointing performance in some of its units by establishing three geographic units, representing UK, France and Other Countries, each headed by a senior executive. These managers were charged with turning their region into a profit centre by strictly controlling costs and focusing on maximizing profits from existing retail space.⁵⁵ In 2015 under new CEO Véronique Laury, and while retaining the three core market areas, Kingfisher launched a new corporate strategy emphasizing greater integration of its five major brands, with a bigger proportion of products sold globally across the brands.⁵⁶ Some years ago, California's Apple Computer reorganized from a functional to a geographical structure to facilitate manufacture and delivery of Apple computers to customers around the world. Exhibit 4.11 contains a partial organization structure illustrating the geographical thrust. At Apple this structure was introduced to focus managers and employees on specific geographical customers and sales targets.

The strengths and weaknesses of a geographic divisional structure are parallel to those of the divisional organization characteristics listed earlier in Exhibit 4.10. The geographic division can adapt to specific conditions of its own country or region, and employees may identify with regional goals rather than with the overall corporate vision. Horizontal coordination within a region may be emphasized rather than linkages across regions or to the head office, and this can detract from overall corporate synergies. This is true both for national firms with regional divisions and for multinational firms with national or regional units around the world.

EXHIBIT 4.11 Geographical Structure for Apple Computer

Source: Apple Computer, Inc. regions of the world, www.apple.com/find/areas.htm, 18 April 2000.

Matrix Structure

Sometimes, a **matrix structure** is developed in an effort to give equal emphasis and attention to product and function, or product and geography. The matrix is more likely to be introduced when, say, technical expertise and product innovation and change are assessed to be of equal importance. When it is assessed that functional, divisional and geographical structures combined with horizontal linkage mechanisms are not working effectively, the matrix may be adopted as a remedy.

The matrix is a strong form of horizontal linkage as, in principle, it enables two objectives to be pursued simultaneously, as shown in Exhibit 4.12. In this structure, product managers and functional managers, for example, have equal authority within the organization, and employees report to both of them. The matrix structure is similar to the use of full-time integrators or product managers described earlier in this chapter (Exhibit 4.3), but within the matrix structure, the product managers (horizontal) are given formal authority equal to that of the functional managers (vertical).

Conditions for the Matrix

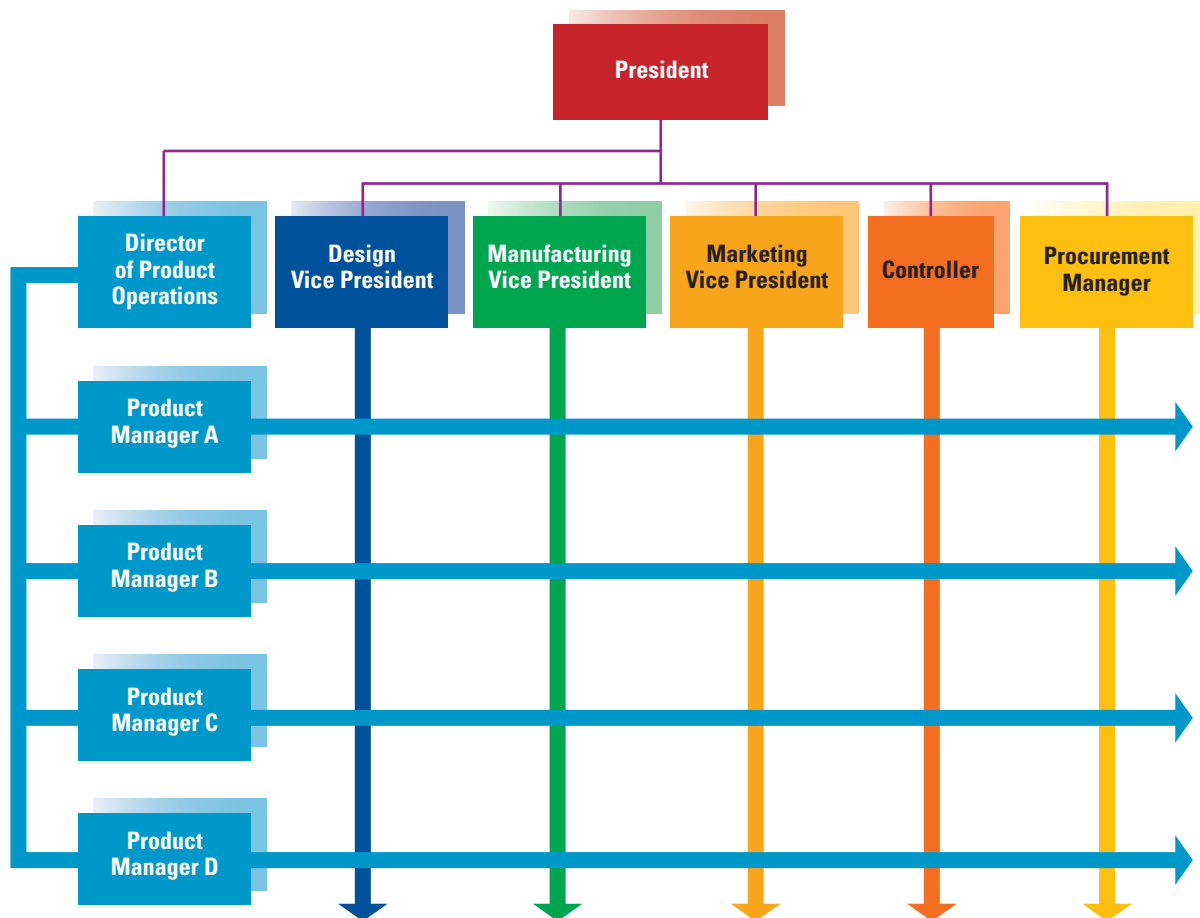
A dual hierarchy may seem an unusual way to design an organization as it contradicts the classical principle of unity of command, but it can be a workable structure, particularly when one or more of the following conditions is met:⁵⁷

- **Condition 1.** Pressure exists to share scarce resources across product lines. The organization is often medium sized and has a moderate number of product lines. There are pressures for the shared and flexible use of people and equipment across the product lines. For example, the organization is not large enough to assign engineers full-time to each product line, so engineers are assigned part-time to several products or projects.

- *Condition 2.* Environmental pressures exist for two or more critical outputs, such as for in-depth technical knowledge (functional structure) and frequent new products (divisional structure). The dual pressure requires sharing of power between the functional and product sides of the organization, and a dual-authority structure is needed to maintain that balance.
- *Condition 3.* The environmental domain of the organization is both complex and uncertain. Frequent external changes and high interdependence between departments require a large amount of coordination and information processing in both vertical and horizontal directions.

In principle, a dual-authority structure can help ensure a balance between vertical and horizontal aspects of organizations.

EXHIBIT 4.12 Dual-Authority Structure in a Matrix Organization



Referring again to Exhibit 4.12, assume the matrix structure is for a clothing manufacturer. Product A is footwear, product B is outerwear, product C is night clothes and so on. Each product line serves a different market and customers. As a medium-sized organization, the company must effectively use people from manufacturing, design and marketing to work on each product line. There are not enough designers to warrant a separate design department for each product line, so the designers are shared across product lines. Moreover, by keeping the manufacturing, design and marketing functions intact, it allows employees to develop the in-depth expertise to serve all product lines efficiently.

The matrix formalizes horizontal teams along with the traditional vertical hierarchy and tries to give equal balance to both. However, the matrix may shift one way or the other. Many companies have found a balanced matrix hard to implement and maintain because one side of the authority structure often dominates. For employees, serving two masters can prove a confusing and excessively demanding experience. As a consequence, two variations of matrix structure have evolved – the **functional matrix** and the **product matrix**. In a functional matrix, the functional bosses have primary authority and the project or product managers simply coordinate product activities. In a product matrix, by contrast, the project or product managers have primary authority, and functional managers simply assign technical personnel to projects and provide advisory expertise as needed. For many organizations, one of these approaches becomes dominant as it accommodates established power relations and reduces conflicts arising from dual lines of authority.⁵⁸

All kinds of organizations have experimented with versions of the matrix, including hospitals, consulting firms, banks, insurance companies, government agencies and many types of industrial firms.⁵⁹ A 2016 McKinsey study of 4000 US-based corporate employees found that the great majority have some kind of matrix reporting relationships.⁶⁰ This structure has been used by large, global organizations such as Unilever and Procter & Gamble, which fine-tune the matrix to suit their own priorities and traditions. Many chemical and pharmaceutical companies operate with various forms of the matrix structure to address the complexities of managing lengthy product portfolios in widely divergent market situations throughout the world. For example, Bayer AG, the German-based multinational, restructured in the mid-1980s from a function-based hierarchy to a three-layered matrix. The company was initially divided into six broad product areas, now streamlined down to the three areas of Healthcare, CropScience and Material-Science. A functional divisional system was also established with functions such as human resources, marketing and finance. Finally, geographic regions were established. The result was 19 interwoven units or divisions that collaborate together in multiple configurations.⁶¹

Strengths and Weaknesses

The matrix structure is often introduced in challenging conditions when an equality of contribution as well as close collaboration is required between, for example, product and functional elements. To be effective, the structure depends on high levels of organizational trust and mutual understanding between managers in different places in the matrix. The dual-authority structure can assist communication and coordination to cope with rapid environmental change and enables an equal balance between product and functional bosses.⁶² The matrix can facilitate discussion and adaptation to unexpected problems. It often works well in organizations of moderate size with a few product lines. Exhibit 4.13 summarizes the strengths and weaknesses of the matrix structure based on what we know of organizations that use it.⁶³

A significant disadvantage of the matrix is that employees sometimes experience dual authority, reporting to two bosses and juggling conflicting demands. This can be frustrating and confusing, especially if roles and responsibilities are not clearly defined by top managers.⁶⁴ Working effectively in a matrix demands excellent interpersonal and conflict-resolution skills. The matrix also forces managers to spend a great deal of time in coordinating meetings.⁶⁵ If managers do not adapt to the information and power-sharing processes required to secure the benefits of the matrix, the system will underperform and fall into disrepute as employees revert to an earlier structure or develop ‘work-arounds’ that shadow and further disrupt the matrix. Managers must collaborate with one another rather than rely on vertical authority in decision-making. An example of a successful matrix design is discussed in *In Practice* CNH Global NV.



ONLINE BRIEF 4.5



ONLINE
COUNTERPOINT 4.4

EXHIBIT 4.13 Strengths and Weaknesses of Matrix Organization Structure

Strengths	Weaknesses
1. Achieves coordination necessary to meet dual demands from customers 2. Flexible sharing of human resources across products 3. Suited to complex decisions and frequent changes in unstable environment 4. Provides opportunity for both functional and product skill development 5. Best in medium-sized organizations with multiple products	1. Causes participants to experience dual authority, which can be frustrating and confusing 2. Means participants need good interpersonal skills and extensive training 3. Is time consuming; involves frequent meetings and conflict resolution sessions 4. Will not work unless participants understand it and adopt collegial rather than vertical type relationships 5. Requires great effort to maintain power balance

Source: Adapted from Robert Duncan, 'What Is the Right Organization Structure? Decision Tree Analysis Provides the Answer', *Organizational Dynamics* (Winter 1979), 429.



IN PRACTICE

CNH Global NV⁶⁶

CNH was formed in the late 1990s through a merger of the construction and agricultural equipment companies Case and New Holland. It is the world's largest company in its market segment. A third of all combine harvesters and backhoe loaders, and a quarter of all tractors and skid steer loaders sold worldwide are built by CNH, a majority owned subsidiary of the Italian vehicle manufacturer Fiat SpA.

CNH managers wanted to benefit from the economies of scale that could be leveraged from such a large company with many common elements in its products, while at the same time retaining the ability to tailor its products to different customer needs and desires in its different markets across the world. It aimed to achieve this through a 'multi-brand, multi-channel' strategy while establishing a global

product development platform. Put simply, this meant producing a variety of lines for different markets from a few large production facilities. In order to succeed there had to be a close, interactive relationship between the marketing and sales teams, on the one hand, who were close to customers, and the production experts, on the other.

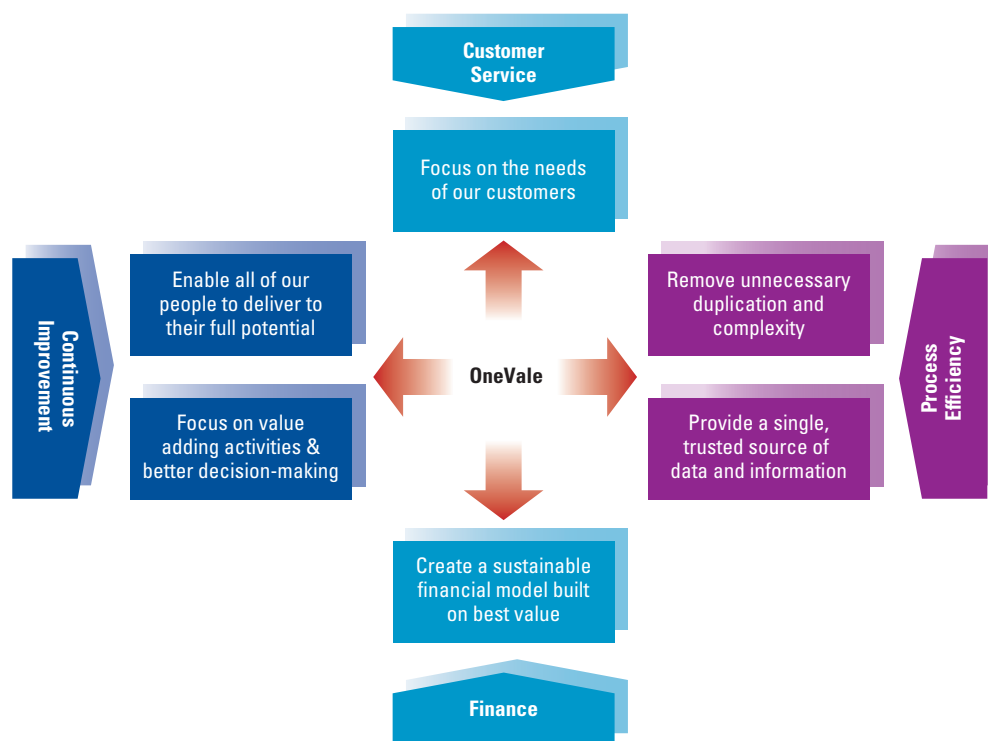
Formerly, CNH put in place matrix accountability to structures to support this interactivity between the product and geographic managers. CNH has been largely successful in its efforts, despite facing challenging market conditions at times. The company is rated highly for employee satisfaction, despite closing several production facilities while moving towards global production platforms. To meet new requirements and industry demands, which are taking place at an ever-accelerating rate, and growing magnitude fuelled by trends such as digitalization, automation, electrification and servitization, CNH needs to adapt, change and revitalize continuously to meet these challenges and successfully generate long-term value. 'We have developed five divisions which will be fully responsible for the global growth and performance of their respective businesses, increasing focus and accountability', said Hubertus Mühlhäuser, Chief Executive Officer, CNH Industrial.

Horizontal Structure

A recent approach to organizing, considered in Chapter 3, is the **horizontal structure**, which organizes employees around core processes. Organizations may be prompted to move towards a horizontal structure as a consequence of an intervention like total quality management or a procedure called re-engineering. **Re-engineering**, or *business process re-engineering*, involves the redesign of a vertical organization along its horizontal workflows and processes. A **process** refers to an organized group of related tasks and activities that work together to transform inputs into outputs that create value for customers.⁶⁷ Re-engineering aspires to change the way managers think about how work is done. Rather than focusing on narrow jobs structured into distinct functional departments, re-engineering emphasizes core processes that cut horizontally across the organization and involve teams of employees working together to serve customers. Examples of processes include order fulfilment, new product development and customer service.

A good illustration of process engineering is the redesign of customer services at the Vale of Glamorgan Council, a medium-sized local government in Wales, UK. The council provides citizens with a range of services ranging from waste disposal to administration of planning and building regulations. In the past, citizens had to reach a specific department in order to make a complaint or arrange an appointment. It was often hard to reach the right person, and both staff and citizens wasted time leaving phone messages back and forth. After studying the problem and getting feedback from service users, the council reorganized the way it does business. Its OneVale initiative established a centralized customer contact centre in conjunction with a computerized information management system. Now, anyone needing to deal with the Council calls the contact centre, whose staff are empowered to deal with simple enquires and complaints themselves, and can set up appointments for all the Council's different departments. The council's phone lines are open late into the evening, and the Vale of Glamorgan has won awards for innovation and customer satisfaction (see Exhibit 4.14).⁶⁸

EXHIBIT 4.14 OneVale Initiative



When an organization is re-engineered to a horizontal structure, all the people throughout the organization who work on a particular process (such as customer service, claims handling or order fulfilment) have better access to one another so they can communicate and coordinate their efforts. The horizontal structure reduces the vertical hierarchy and erodes old departmental boundaries. This approach is generally facilitated by the use of information and communication technologies as well as by changing attitudes to work flexibility that have occurred in the workplace and the business environment over the past 25 years (see Counterpoint 4.4). Computerization and internet connectivity permits wide-ranging integration and coordination. Customers expect faster and better service, and employees may have the opportunity to develop new skills and assume more responsibility. Organizations mired in a vertical mindset where each function or department operates within a separate 'silo' have a hard time meeting these challenges. Thus, numerous organizations have experimented with horizontal mechanisms such as cross-functional teams to achieve coordination across departments or task forces to accomplish temporary projects. Increasingly, organizations are shifting away from hierarchical, function-based structures to structures based on horizontal processes.

COUNTERPOINT 4.4

It would be misleading to suggest that such changes are easily or smoothly introduced, or that they are always welcomed by customers or employers. Contacting call centres can be frustrating and unsatisfactory for customers and employees alike. Change is often driven by the objective of making savings rather than improving service. Or, service improvements are anticipated but an emphasis on cost reduction or a lack of vision results in poor service and disenchanted employees. Call centres can be 'designed' in various ways – at worst, there are those that mirror highly fragmented organizations, where callers wait in queues without any information and are then passed around to different employees who never give their name; and at best there are those that deal with calls promptly, courteously and are trained to manage the process – that is, take care of the handling of each call or 'case' from initial contact to its resolution.



Characteristics

An illustration of a company re-engineered into a horizontal structure appears in Exhibit 4.15. Such an organization has the following characteristics:⁶⁹

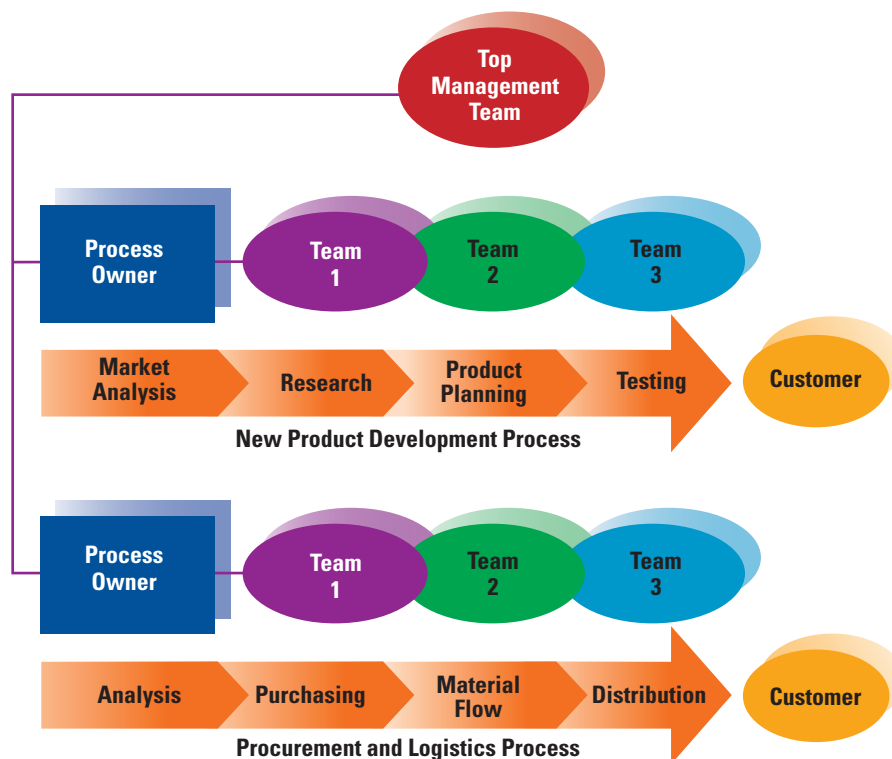
- ONLINE BRIEF 4.6** ■ Structure is created around cross-functional core processes rather than tasks, functions or geography. Thus, boundaries between departments are obliterated.
- Self-directed teams, not individuals, are the basis of organizational design and performance.
 - Process owners have responsibility for each core process in its entirety. For example, until the 1990s, most Britons bought houses through building societies, a kind of financial cooperative. Banks were just not set up to smoothly handle all the different elements that go in to mortgage approval. Prospective buyers had to visit several different departments of the bank to arrange the loan, arrange the legal paperwork, obtain mortgage insurance etc., with delays of weeks if not months. As the property market became more competitive, Barclays Bank realized that if it wanted to build its mortgage business, it had to offer a comprehensive, one-stop service. It brought together all the different expertise and signing authority into a single department, allowing the process to be greatly streamlined. By 2014 the average time for Barclays to provide an offer letter was down to nine days, despite considerably tightened government requirements in the wake of the global financial crisis.⁷⁰ Barclays crystallized this new business process into its Home Finance Division. Both approvals and, crucially, profits, rose substantially, and by 2014 Barclays consolidated its position in the top five mortgage lenders in the UK.⁷¹

- People on the team are given the skills, tools, motivation and authority to make decisions central to the team's performance. Team members are cross-trained to perform one another's jobs, and the combined skills are sufficient to complete a major organizational task.
- Teams have the freedom to think creatively and respond flexibly to new challenges that arise.
- Customers drive the horizontal corporation. Effectiveness is measured by end-of-process performance objectives, based on the goal of bringing value to the company's stakeholders including customers and investors.
- The culture is one of openness, trust and collaboration, focused on continuous improvement. The culture values employee empowerment, responsibility and well-being.



Avaya Ireland, a hi-tech telecommunications firm, is featured in the *In Practice* box which follows. Avaya successfully shifted to a flexible, less hierarchical model in order to meet the demands of diverse customers and a rapidly changing product mix.

EXHIBIT 4.15 A Horizontal Structure



Source: Based on Frank Ostroff, *The Horizontal Organization* (New York: Oxford University Press, 1999); John A. Byrne, 'The Horizontal Corporation', *BusinessWeek* (December 20, 1993), 76–81; and Thomas A. Stewart, 'The Search for the Organization of Tomorrow', *Fortune* (May 18, 1992), 92–98.

Strengths and Weaknesses

As with all structures, the horizontal structure has weaknesses as well as strengths. Some of the main strengths and weaknesses of the horizontal structure are listed in Exhibit 4.16.

EXHIBIT 4.16 Strengths and Weaknesses of Horizontal Structure

Strengths	Weaknesses
1. Promotes flexibility and rapid response to changes in customer needs 2. Directs the attention of everyone towards the production and delivery of value to the customer 3. Each employee has a broader view of organizational goals 4. Promotes a focus on teamwork and collaboration 5. Improves quality of life for employees by offering them the opportunity to share responsibility, make decisions and be accountable for outcomes	1. Determining core processes is difficult and time consuming 2. Requires changes in culture, job design, management philosophy, and information and reward systems 3. Traditional managers may balk when they have to give up power and authority 4. Requires significant training of employees to work effectively in a horizontal team environment 5. Can limit in-depth skill development

Sources: Based on Frank Ostroff, *The Horizontal Organization: What the Organization of the Future Looks Like and How It Delivers Value to Customers* (New York: Oxford University Press, 1999); and Richard L. Daft, *Organization Theory and Design*, 6th ed. (Cincinnati, OH: South-Western, 1998), 253.



IN PRACTICE

Avaya Ireland

Avaya Ireland is part of the global Avaya corporation, one of the many offspring of the giant AT&T telecommunications business that was broken up in the 1980s into smaller, more flexible and more entrepreneurial entities. Avaya specializes in the manufacture of networking components, key building blocks of today's information society. Ireland is one of the main European centres for the computer industry, so Avaya Ireland had a dual advantage of location and growing demand for its product sector. Despite this positive market situation, the company languished through the 1980s and early 1990s. The company was run by a rigid functional hierarchy, with poor cross-functional communication. Costs were high compared with competitors and other subsidiaries of the Avaya group, and staff morale was poor. Avaya Ireland was in real danger of closing. In the mid-1990s top managers realized that they had to adapt to the times or the company would die. A management steering group was established to animate a process of internal dialogue, called Engaging our People. The

highly inclusive, and often difficult process identified four main areas of weakness: communications, training, the manager/employee relationship and lack of teamwork. An action plan, based on the Business Excellence Model⁷² was developed. Structural changes to realize the company's vision entailed replacing the rigid organizational hierarchy with cross-functional teams focused on core products, from conception, through production to end-user support. Armed with this new focus, the company was able to contain costs while improving product quality. Avaya invited customers to visit the plant and meet its new product-focused cross-functional teams. This was a highly successful initiative – customers saw that Avaya had an open culture and knowledgeable staff. Satisfaction with the company's networking products soared. In 1998 Avaya won the Irish Business Excellence overall award, and two years later a prize at the European Quality Awards as well as becoming one of the first Irish companies to be awarded ISO9001 certification. The change-driving steering group remains in place, and Avaya Ireland continues to win high marks for customer service and innovation, ranking in the Thompson Reuters top 100 for innovation for three years to 2013.⁷³ The company has now moved strongly into the related high-growth area of voice over internet protocol (VOIP) telephony.⁷⁴ In 2020, the main innovation project is helping customers become even more digital and accelerating digital transformation.

The most significant strength of the horizontal structure is that it can dramatically improve flexibility and responsiveness. The structure tends to direct attention towards the customer, leading to greater customer satisfaction as well as improvements in productivity, speed and efficiency. In addition, because there are no boundaries between functional departments, employees can take a broader view of organizational goals rather than being focused on the goals of a single department. The horizontal structure promotes an emphasis on teamwork and cooperation, so that team members share a commitment to meeting common objectives. Finally, the horizontal structure can improve the quality of life for employees by giving them opportunities to share responsibility, make decisions and contribute significantly to the organization.

A potential weakness of the horizontal structure is that it can harm rather than help organizational performance, unless managers carefully determine which core processes are critical for bringing value to customers. Simply defining the processes around which to organize can be difficult. In addition, shifting to a horizontal structure requires significant changes in culture, job design, management philosophy and information and reward systems, and can therefore be complicated, time consuming and thus costly. Traditional managers may balk at having to share elements of power and authority, and find it difficult to learn to exercise leadership through team facilitation. Employees have to be trained to work effectively in a team environment, and they may be suspicious of the company's motives in dismantling overt hierarchy or hostile to its disruptive impact upon valued routines. Finally, because of the cross-functional nature of work, a horizontal structure can limit in-depth knowledge and skill development unless measures are taken to enable employees opportunities to maintain and build technical expertise.

Virtual Network Structure

The virtual network structure extends the concept of horizontal coordination and collaboration beyond the boundaries of the traditional organization. Many of today's organizations farm out some of their activities to other companies. **Outsourcing**, which means the contracting out of aspects of work (e.g. manufacturing, information technology or credit processing) to other companies, is a significant trend in many industries that has implications for organization structure.⁷⁵ Accenture, for example, handles all aspects of information technology for the British food retailer J. Sainsbury's. Companies in countries such as India and Malaysia, as well as European sites such as Scotland and Eastern Europe, manage call centre and technical support for multinational corporations including financial companies, computer vendors and mobile phone companies. Entire chunks of aeroplanes manufactured by Canada's Bombardier and Brazil's Embraer are engineered and built by outside contractors, often outside Canada and Brazil respectively. Fiat Auto is involved in multiple complex outsourcing relationships with other companies handling logistics, maintenance and the manufacturing of some parts.⁷⁶

These interorganizational relationships reflect a significant shift in organization design. An increasing number of organizations take outsourcing to the extreme and create a virtual network structure. With a **virtual network structure**, sometimes called a *modular structure*, the firm subcontracts many or most of its major processes to separate companies and coordinates their activities from a small headquarters organization. Nowadays fashionwear companies like Nike frequently produce none of their own clothing, focusing instead on building brand value through comprehensive marketing efforts and exercising tight control over their suppliers. The virtual network structure has become much more feasible as a result of advances in information and communication technologies.⁷⁷

How the Structure Works

The virtual network organization may be viewed as a central hub surrounded by a network of outside specialists. Rather than being housed under one roof or located within one organization, services such as accounting, design, manufacturing, marketing and distribution are outsourced to separate companies that are connected electronically to a central office. Organizational partners located in different parts of the world may use secure networking to exchange

data and information so rapidly and smoothly that a loosely connected network of suppliers, manufacturers and distributors can look and act like one seamless company. The virtual network form can also incorporate a free-market style to replace the traditional vertical hierarchy. In this case, contractors can be brought into and released from the system to meet changing needs.

With a network structure, the hub maintains control over processes in which it has difficult-to-imitate capabilities and transfers other activities – along with the decision-making for, and operational control over, those activities – to other organizations who are specialists. These organizations organize and accomplish their work using their own ideas, assets and tools.⁷⁸ The idea is that a firm can concentrate on its ‘core competence’ – what is key to its survival – and contract out everything else to companies with distinctive competence in those specific areas, thereby in principle enabling the organization to do more with less.⁷⁹ The network structure is often advantageous in reducing costs for start-up companies, such as Dicole, a Finnish IT firm that works with companies to implement Web 2.0 technologies, featured in the next *In Practice* box below.



IN PRACTICE

Leveraging Web 2.0

Virtual networking is made possible by new technologies, and the ever-expanding universe of virtual communication creates new opportunities almost daily. Companies small and large are exploring ways to build virtual networks and strengthen collaboration across the globe, using the boundless potential of Web 2.0, and particularly new thinking in virtual communications. This *In Practice* discusses several companies' and organizations' approaches to Web 2.0.

LinkedIn was founded in late 2002 by Reid Hoffmann when he recruited a team of former colleagues from SocialNet and PayPal to work on a new idea. The idea was a site where employers post jobs and job seekers post CVs. In 2015, most of the company's revenue came from selling access to information about its members to recruiters and sales professionals. As of June 2019, LinkedIn had 630 million registered members in 200 countries.

One of the simplest forms of virtual communication is the Wiki, a word that comes from the Hawaiian for ‘fast’, and involves working collaboratively and in real time on a single text. Wikipedia, the online

collaborative encyclopedia, is the best known example of wiki technology, but wikis are used increasingly within companies for internal communication, and increasingly for technical documentation of work processes. As employees become more familiar with wikis, they are being used to develop product manuals and instructions for end users outside the company. Similarly, the same technology used to create YouTube, the popular site for uploading, viewing and sharing video clips, can be used to transfer knowledge within a company quickly and cheaply. Large companies such as IBM use technology similar to that underpinning the social networking site Facebook to help employees to get to know and understand each other – over 400,000 IBM employees have their own pages on the company's ‘Blue Pages’, which can include photos, employees' places in the IBM hierarchy, contact information and even their own weblogs.

While many companies use versions of wikis, blogs, LinkedIn and Facebook, some companies work with their own customer organizations to combine the various technologies together in a tailor-made approach to virtual network communications. One such example is Finnish company Dicole's Knowledge Work Environment.⁸⁰

The concept can be taken even further through the product development technique called *crowdsourcing*, where social networking technologies are used to develop new products. Many different approaches have been tried. One example is Threadless (www.threadless.com),

a Chicago-headquartered t-shirt company that invites anyone to submit t-shirt designs. If the design is selected, the designer is paid an upfront fee and a commission depending on the number of t-shirts sold with that design. By 2013, Threadless had accepted 264,055 designs from 1443 artists, and had a total of 2,423,041 'community members'

(customers) worldwide. Threadless's first Chief Technology Officer, Harper Reed, went on to run technology for Barack Obama's successful 2012 US presidential campaign, and joined Paypal to head their software development division in 2015. Reed is not shy, either; he describes himself on his website as 'Probably one of the coolest guys ever'.^{81, 82}

Strengths and Weaknesses

Exhibit 4.17 summarizes the strengths and weaknesses of the virtual network structure. One of the major strengths is that even quite small organizations can be truly global, drawing on resources worldwide to achieve the best quality and price and then selling products or services worldwide just as easily through subcontractors. The network structure also enables new or small companies to develop products or services and get them to market rapidly without huge investments in factories, equipment, warehouses or distribution facilities. The ability to arrange and rearrange resources to meet changing needs and best serve customers gives the network structure flexibility and rapid response. New technologies can be developed quickly by tapping into a worldwide network of experts. The organization can continually redefine itself to meet changing product or market opportunities. A final strength is reduced administrative overhead. Large teams of staff specialists and administrators are not needed. Managerial and technical talent can be focused on key activities that provide competitive advantage while other activities are outsourced.⁸³



ONLINE BRIEF 4.7

EXHIBIT 4.17 Strengths and Weaknesses of Virtual Network Structure

Strengths	Weaknesses
1. Enables even small organizations to obtain talent and resources worldwide 2. Gives a company immediate scale and reach without huge investments in factories, equipment or distribution facilities 3. Enables the organization to be highly flexible and responsive to changing needs 4. Reduces administrative overhead costs	1. Managers do not have hands-on control over many activities and employees 2. Requires a great deal of time to manage relationships and potential conflicts with contract partners 3. There is a risk of organizational failure if a partner fails to deliver or goes out of business 4. Employee loyalty and corporate culture might be weak because employees feel they can be replaced by contract services

Sources: Based on Linda S. Ackerman, 'Transition Management: An In-Depth Look at Managing Complex Change', *Organizational Dynamics* (Summer 1982), 46–66; and Frank Ostroff, *The Horizontal Organization* (New York: Oxford University Press, 1999), fig 2.1, 34.

The virtual network structure also has a number of weaknesses.⁸⁴ The primary weakness is a potential lack of control associated with taking decentralization to the extreme. Managers do not have all operations under their jurisdiction and must rely on contracts, coordination and negotiation to hold things together. This can also mean increased time spent managing relationships

with partners and resolving conflicts, which can be especially complex if partners are operating in different continents, many time zones away. Communications technologies have greatly reduced effective distance, but there is no substitute for face-to-face negotiations when complex, serious issues arise.

A problem of equal importance is the risk of failure if one organizational partner fails to deliver, has a plant burn down or goes out of business. Managers in the headquarters organization have to act quickly to spot problems and find new arrangements. Finally, from a human resource perspective, employee loyalty can be weak in a network organization, both because of concerns over job security and because developing a cohesive corporate culture is difficult between far-flung units or outsourced functions. Employees may feel that they could be replaced at any time by contract services. Turnover of staff may be higher because emotional commitment between the organization and employees is low. With changing products, markets and partners, the organization may need frequently to reshuffle employees to get the correct mix of skills and capabilities.

Hybrid Structure

As a practical matter, many structures in the real world do not exist in the pure forms we have outlined in this chapter. Organizations often use a **hybrid structure** that combines characteristics of various approaches tailored to specific strategic needs. Most companies combine characteristics of functional, divisional, geographical, horizontal or network structures to take account of the relative strengths and weaknesses of these structures in their own particular business.

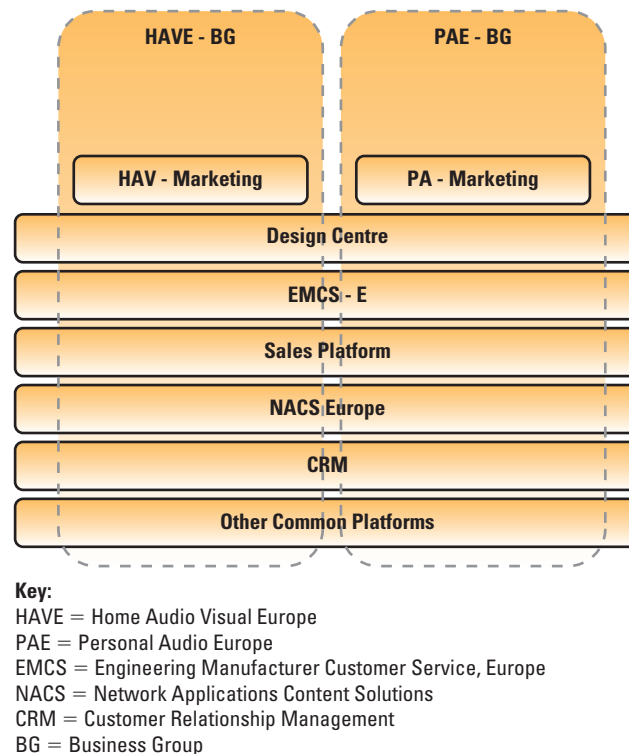


ONLINE BRIEF 4.8

One type of hybrid that is often used is to combine characteristics of the functional and divisional structures. When a corporation grows large and has several products or markets, it typically is organized into self-contained divisions of some type. Functions that are important to each product or market are decentralized to the self-contained units. However, some functions that are relatively stable and require economies of scale and in-depth specialization are also centralized at headquarters.

Sony Europa is the European arm of the Japanese electronics giant, Sony Corporation. The company has been active in Europe since 1960 and has enjoyed success across the continent due to the famed high quality of Sony products. However, the company's involvement in Europe developed in an unplanned way, with marketing offices and manufacturing plants growing up in many different countries. Country managers, by default, became leaders of mini-empires, and there was little cross-European coordination. The weaknesses of this approach could be overlooked while Sony was growing, but as the company faced stiffer competition on both price and quality from the 1990s, company leaders realized a more strategic and coordinated approach to organizational structure was needed. Successive efforts by Sony's top management to introduce some coordination resulted in the establishment of business units that grouped together consumer electronics, business and professional products, and OEM products geared to industrial purchasers. While this improved coordination, each division had its own functional units such as HR, marketing and logistics, detracting from possible cost synergies and overall corporate identity. In 2002, Sony Europa's new president Mike Tsurumi established a hybrid structure which retains product divisions, but centralizes the functional areas of HR, Finance, Sales, Marketing and Infrastructure⁸⁵ (see Exhibit 4.18 as applied to Marketing). A hybrid structure such as this is often preferred over the pure functional, divisional, horizontal or virtual network structure because it can provide some of the advantages of each and overcome some of the disadvantages.

EXHIBIT 4.18 New Structure of European Marketing Introduced During the Era of Mike Tsurumi



Prof. Paul Sparrow: www.lums.lancs.ac.uk/files/hr/10044.pdf

Applications of Structural Design

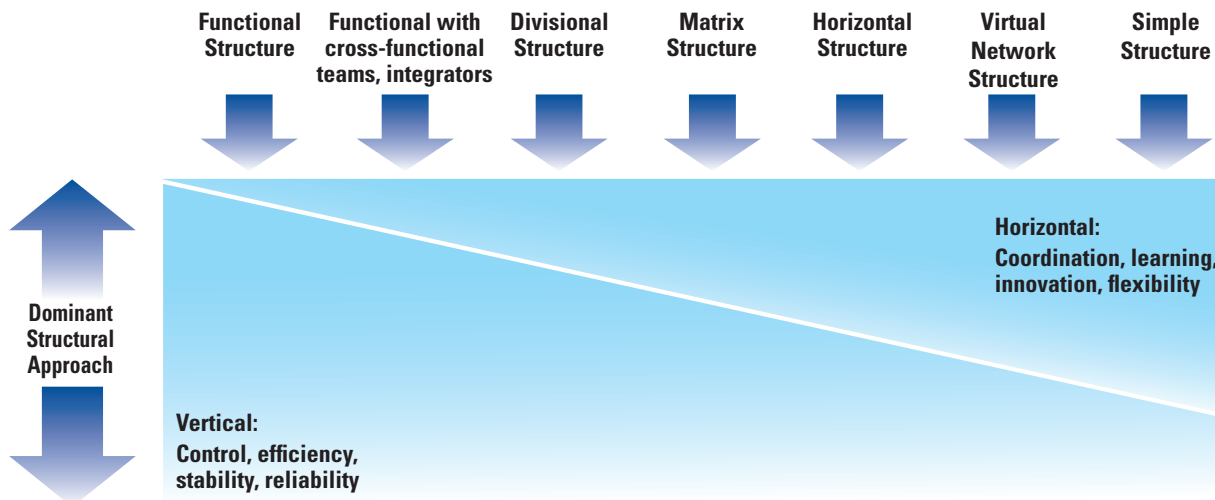
Each type of structure is applied in different situations and is intended to address different demands. In describing the various structures, we touched briefly on conditions such as environmental stability or change and organizational size that are related to structure. Each form of structure – functional, divisional, matrix, horizontal, network, hybrid – represents a tool that can make an organization more or less effective, depending on the exigencies of the situation and the capacity of the structure to respond effectively to their often conflicting demands.

Structural Alignment

One of the most important roles of top managers in structural design is finding the right balance between vertical control and horizontal coordination, depending on the constantly changing needs of the organization. Vertical control tends to be associated with goals of efficiency and stability, while horizontal coordination is associated with learning, innovation and flexibility. Exhibit 4.19 shows a simplified continuum that illustrates how structural approaches are associated with vertical control versus horizontal coordination. The functional structure is most appropriate when managers believe that efficiency in meeting organizational goals can be achieved through coordination by vertical hierarchy. However, the narrow focus on specific outcomes that vertical hierarchy tends to produce can be insufficiently flexible to account for rapidly changing pressures in a complex business environment.

At the opposite end of the scale, horizontal structures are more appropriate when innovation is a priority. In this case, continuous coordination among functions fosters organizational learning. The horizontal structure enables organizations to differentiate themselves and respond quickly to changes, but often at the expense of short-term efficiency. The virtual network structure offers even greater flexibility and potential for rapid response by allowing the organization to add or subtract pieces as needed to adapt and meet changing demands and challenges, as well as searching the world for lower costs and increased productivity. The simple structure to the right is the type of organization with the greatest flexibility because the founder and the employees work together informally. The employees gain experience from the way they handle customers individually and by the many different demands from customers. Exhibit 4.19 also shows how other types of structure defined in this chapter – functional with horizontal linkages, divisional and matrix – represent intermediate steps on the organization's path to cost-efficiency, innovation and learning. The exhibit does not include all possible structures, but it illustrates how organizations attempt to balance competing priorities and demands (see Counterpoint 4.5).

EXHIBIT 4.19 Relationship of Structure to Organization's Need for Efficiency versus Learning



COUNTERPOINT 4.5

In this chapter we have presented a range of formal schemas of design. In practice, however, organizations are ongoing projects in which elements and remnants of different structures are overlapped and overlaid. The result is more of a patchwork than a 'hybrid' that is the product of successive efforts to mitigate for the limitations or weaknesses ascribed to a previous structure. Neither should it be overlooked that structures are generally imposed from the top, sometimes mainly to signal the arrival of a new CEO or management team with 'fresh ideas', and they then encounter resistance which moderates their impact and (in)effectiveness.

Symptoms of Structural Deficiency

Top executives periodically evaluate organization structure to determine whether it is appropriate to changing conditions and aspirations. Many organizations try one organization structure and then reorganize to another structure in an effort to develop a better fit between executive aspirations, internal reporting relationships and the external environment.

The following are **symptomatic of structural deficiency** (see Counterpoint 4.6).⁸⁶

- *Decision-making is delayed or lacking in quality.* Decision-makers may be overloaded because the hierarchy funnels too many problems and decisions to them. Delegation to lower levels may be insufficient. Another cause of poor-quality decisions is that information may not reach the correct people. Information linkages in either the vertical or horizontal direction may be inadequate to ensure decision quality.
- *The organization does not respond innovatively to a changing environment.* One reason for lack of innovation is that departments are not coordinated horizontally. The identification of customer needs by the marketing department and the identification of technological developments in the research department must be coordinated. Organization structure also has to specify departmental responsibilities that include environmental scanning and innovation.
- *Employee performance declines and goals are not being met.* Employee performance may decline because the structure doesn't provide clear goals, responsibilities and mechanisms for coordination. The structure should reflect the complexity of the market environment and be straightforward enough for employees to effectively work within it.
- *Too much conflict is evident.* In principle, organization structure is intended to enable departmental goals to combine into a single set of goals for the entire organization. When departments act at cross-purposes or departmental priorities are out of alignment with those attributed to the wider organization, it may be that the structure has incorporated insufficient horizontal linkages.

COUNTERPOINT 4.6

Although these problems may arise from 'structural deficiency' that can be corrected by the technical fix of structural redesign, the problems may also be endemic to capitalist work organizations. Decision-making may be delayed because senior managers anticipate that their jobs or career prospects are threatened by change. Innovation may be slow because it poses a threat to established 'empires' within the organization. Declines in employee performance may be associated with frustration and dissatisfaction in, ultimately, being treated as a disposable commodity or, at least, with insufficient dignity as changes are introduced (or imposed) without genuine consultation or adequate understanding of their counterproductive effects. Conflicts between individuals, groups, departments and divisions may arise from the basic understanding that responding to the demands of shareholders and/or customers is not necessarily compatible with improving the wages and conditions (e.g. pensions) of particular groups of employees.

Summary and Interpretation

Organization structure is intended to accomplish two things. It seeks to provide a framework of responsibilities, reporting relationships and groupings, and it is intended to provide mechanisms for linking and coordinating organizational elements into a coherent whole.

Managers make a strategic choice as they orient towards a traditional organization design which emphasizes vertical linkages such as hierarchy, rules and plans, or towards a contemporary learning organization which emphasizes horizontal linkages through cross-functional information systems, direct contact between managers across department lines, temporary task forces, full-time integrators and teams.

Alternatives for grouping employees and departments into overall structural design include functional grouping, divisional grouping, multifocused grouping, horizontal grouping and network grouping. With functional and divisional structures, managers also use horizontal linkage mechanisms to complement the vertical dimension and achieve integration of departments and levels into an organizational whole. With a horizontal structure, activities are organized horizontally

around core work processes. A virtual network structure extends the concept of horizontal coordination and collaboration beyond the boundaries of the organization. Core activities are performed by a central hub while other functions and activities are outsourced to contract partners. The matrix structure attempts to achieve an equal balance between the vertical and horizontal dimensions of structure. Organizations rarely exist in these pure forms. Instead, hybrid or patchwork varieties evolve that incorporate more or less coherent blends of organizing activity that can be analyzed in terms of the various structural types examined in this chapter.

KEY CONCEPTS

centralized	horizontal grouping	organization structure	task force
decentralized	horizontal linkage	outsourcing	teams
departmental grouping	horizontal structure	process	vertical information
divisional grouping	hybrid structure	product matrix	system
divisional structure	integrator	re-engineering	vertical linkages
functional grouping	liaison role	simple structure	virtual network grouping
functional matrix	matrix structure	symptoms of structural	virtual network structure
functional structure	multifocused grouping	deficiency	virtual team

Discussion Questions

- 1 What is the definition of *organization structure*? Does organization structure appear on the organization chart? Explain.
- 2 How do rules and plans help an organization achieve vertical integration?
- 3 When is a functional structure preferable to a divisional structure?
- 4 Large corporations tend to use hybrid structures. Why?
- 5 What are the primary differences between a traditional organization designed for efficiency and a more contemporary organization designed for learning?
- 6 What is the difference between a task force and a team? Between a liaison role and an integrating role? Which of these provides the greatest amount of horizontal coordination?
- 7 What conditions usually have to be present before an organization should adopt a matrix structure?
- 8 The manager of a consumer products firm said, 'We use the brand manager position to train future executives'. Do you think the brand manager position is a good training ground? Discuss.
- 9 Why do companies using a horizontal structure have cultures that emphasize openness, employee empowerment and responsibility? What do you think a manager's job would be like in a horizontally organized company?
- 10 Describe the virtual network structure. Why do you think this is becoming a good structural alternative for some of today's organizations?

Chapter 4 Workbook You and Organization Structure

To better understand the importance of organization structure in your life, do the following assignment. Select one of the following situations to organize:

- A copy and printing shop
- A travel agency
- A sports rental (such as skis or snowboards) in a resort area
- A bakery.

- 1 Write down the mission or purpose of the organization in a few sentences.
- 2 What are the specific tasks to be completed to accomplish the mission?
- 3 Based on the specifics in number 2, develop an organization chart. Each position in the chart will perform a specific task or is responsible for a certain outcome.
- 4 You are into your third year of operation, and your business has been very successful. You want to add a second location a few kilometres away. What issues will you face running the business at two locations? Draw an organization chart that includes the two business locations.
- 5 Five more years go by and the business has grown to five locations in two cities in the same country. How do you keep in touch with it all? What issues of control and coordination have arisen? Draw an up-to-date organization chart and explain your rationale for it.
- 6 Twenty years later you have 75 business locations in 5 European countries. What are the issues and problems that have to be dealt with through organizational structure? Draw an organization chart for this organization, indicating such factors as who is responsible for customer satisfaction, how you will know if customer needs are met and how information will flow within the organization.

Adapted by Dorothy Marcic from 'Organizing', in Donald D. White and H. William Vroman, *Action in Organizations*, 2nd ed. (Boston: Allyn and Bacon, 1982), 154; and Cheryl Harvey and Kim Morouney, 'Organization Structure and Design: The Club Ed Exercise', *Journal of Management Education* (June 1985), 425–429.

CASE FOR ANALYSIS 4.1

Organization Structures

Scotia University

Scotia University is located in Scotland and, like most organizations, periodically restructures its organization to better align its structure to its strategy. Universities are no different from any other organization but tend to use different terminology when it comes to organization structures. Senate is the senior academic body of the university. Legally and constitutionally it is responsible for the academic activity of the university which is teaching and research. Senate is also responsible for student conduct. The University Court is involved in the development of, and ultimately approves, the university's strategic plan. It approves any major financial decisions, such as whether or not to borrow money; it also approves the university's proposed budget.

The principal divisions are normally termed colleges, faculties or schools, with colleges being an overarching structure with direct responsibility to the University Executive or Senior Management Team. The Scotia University College of Arts contains: the School of Humanities; the School of Critical Studies;

the School of Culture & Creative Arts; the School of Modern Languages & Cultures; and its graduate school. The College of Medical, Veterinary and Life Sciences contains: the School of Life Sciences; the School of Medicine; the School of Veterinary Medicine; its graduate school and seven research institutes. The College of Science and Engineering contains: the School of Chemistry; the School of Computing Science; the School of Engineering; the School of Geographical & Earth Sciences; the School of Mathematics & Statistics; the School of Physics & Astronomy; the School of Psychology; and its graduate school. The College of Social Sciences contains: the Business School; the School of Education; the School of Interdisciplinary Studies; the School of Law; the School of Social & Political Sciences; and its graduate school.

The chief executive of a Scottish university has the title Principal and Vice-Chancellor and the Chancellor is a lay-member of the university (not an employee). The Senior Management Group advises the Principal as chief executive officer of the University on matters of policy. It also advises Court and Senate on matters of strategic policy (academic and resource), and acts on a day-to-day basis to implement the policies of Court and Senate. The Senior Management Group (SMG) comprises: the Principal, Vice-Principals, Heads of Colleges, Secretary

of Court and the Director of Finance. The SMG has responsibility for ensuring that the university policies are effectively implemented and that this is done in a coordinated way across the Colleges and University Services. It has a role in developing corporate initiatives and policies, for consideration by Senate and Court. SMG maintains a watching brief on the university strategy and in particular on progress against the strategic key performance indicators (KPIs) and it takes action as required to ensure that the university is positioned to meet them. SMG also has an ongoing responsibility to monitor, assess and address risk, and in liaison with Court, update the strategic risk list annually.

The Court is the governing body of the university, and is sometimes compared to the board of directors of a company. Its powers have been defined over a number of years and are set out in a series of Acts of Parliament, the Universities (Scotland) Acts 1858–1966, and subsidiary regulations and ordinances. The Court has ultimate responsibility for the deployment of resources in the university and for the strategic plans of the institution. It also has a monitoring role in relation to the overall performance of the university, and it holds the Principal accountable for the effective and efficient management of the university. It is responsible for the well-being of staff. With the Senate, it is responsible for the well-being

of students and for the reputation of the university. The Rector is elected by the students of the university to represent their interests. He or she is also, ex-officio, the Chairperson of the University Court, the body which administers the resources of the university. The Rector is not active in university strategy or policy-making. The role is principally as spokesperson and representative for student issues. The Rector's participation in events is entirely voluntary and depends on availability and personal choice.

Case Study Questions

- 1 Describe and explain the structure of Scotia University in terms of its vertical structure and illustrate with an organization chart of the Colleges.
- 2 How is coordination achieved in academic matters?
- 3 Matrix structures are often used to improve horizontal coordination and information sharing, and are ideal structures for a research institute which brings together academics from across the university. Using the College of Social Sciences as your example, illustrate how a matrix organization structure could be used for a research institute in management with two research themes: business management and public management.

Notes

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